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Google Cloud Fundamentals



Prereqs for Course

- No previous experience with Google Cloud
- Some exposure to working on the cloud recommended
- Basic understanding of deploying software on-premises

Prereqs for Hands-on Demos

- Create a free Google Cloud account
- https://console.cloud.google.com/
- Enable billing on that account
- Please watch the getting set up video linked here:
- https://drive.google.com/drive/folders/130rcJUmsy4LANX-7iWasu7KmuvFULkSf?usp=sharing
- All resources available here: https://github.com/janani-ravi-loony/google-cloud-fundamentals

Introductions

I have experience with the Google Cloud Platform:

1. No experience at all
2. 0-1 years of experience
3. 2-3 years of experience
4. 3+ years of experience



Introductions

I have worked on other cloud platforms:

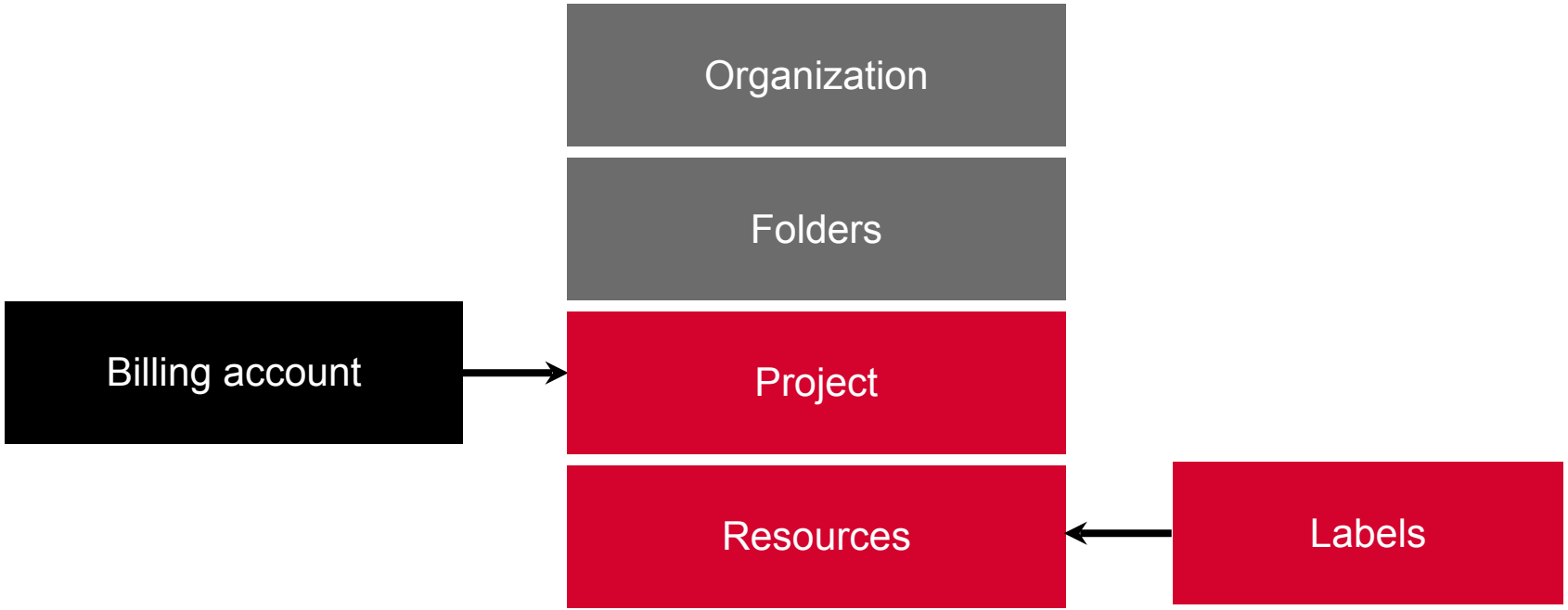
1. Mostly AWS
2. Mostly Azure
3. Other cloud platforms



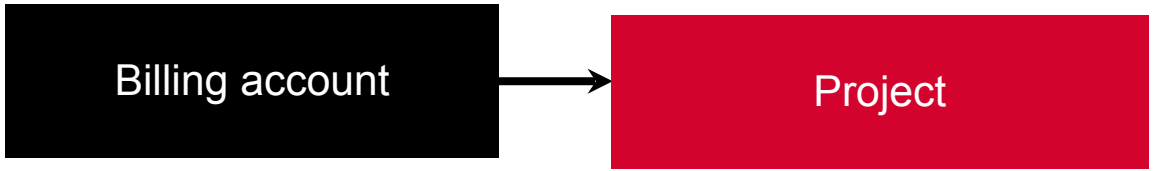


Basic Structure of Resources on the Google Cloud

Resource Hierarchy of Components



Billing Accounts Are Associated with Projects



Labels Are Applied to Resources

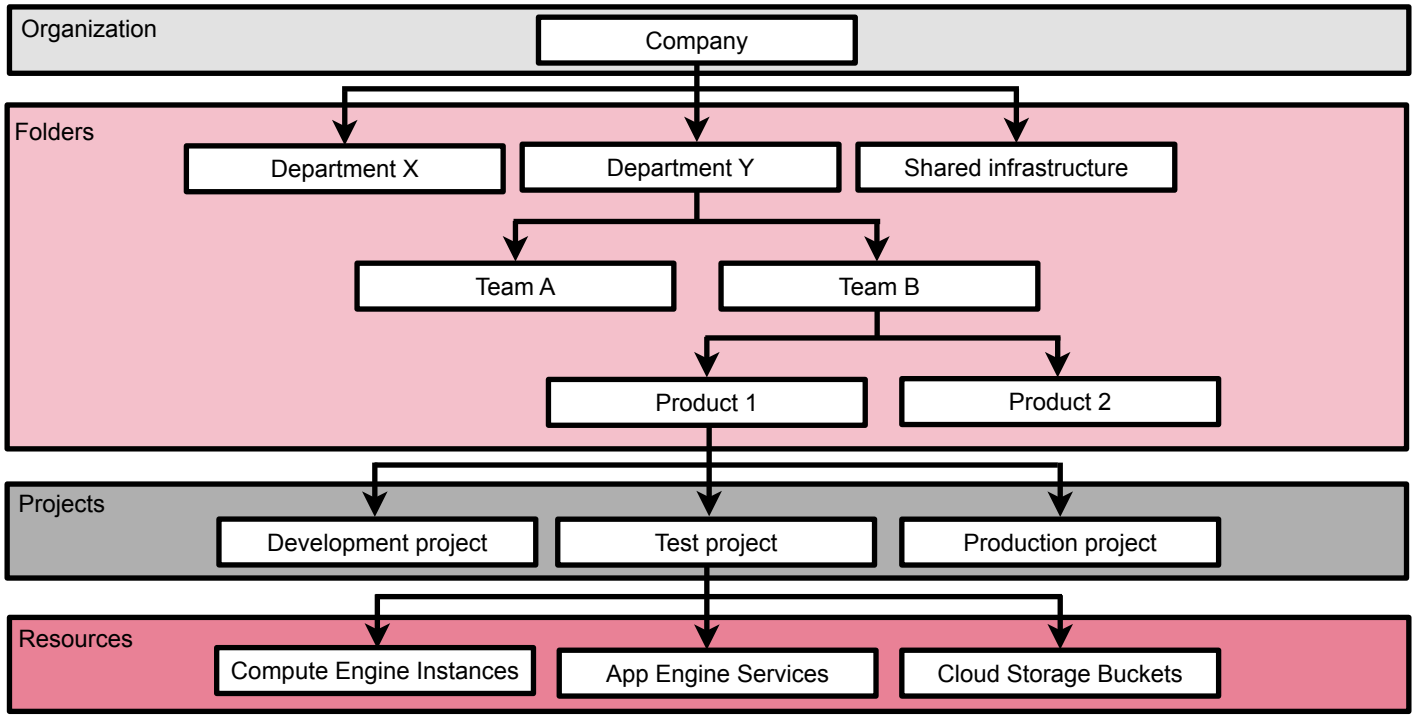


Labels Help in Allocating Costs

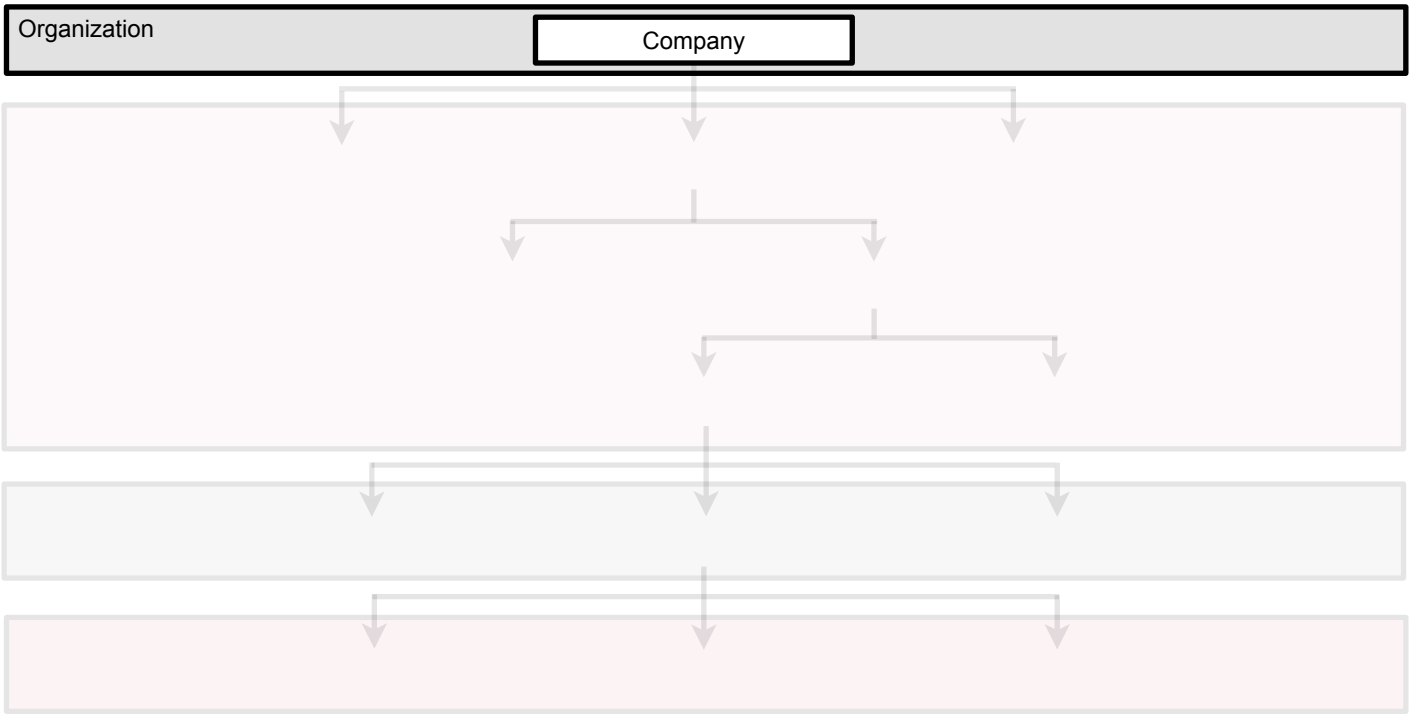
- Categorize resources
  - Different environments
  - Different projects
- Label resources accordingly
  - *env=dev, env=prod*
  - *service=search, service=catalog*
- Can export billing to BigQuery and analyze costs using labels



Google Cloud Hierarchy

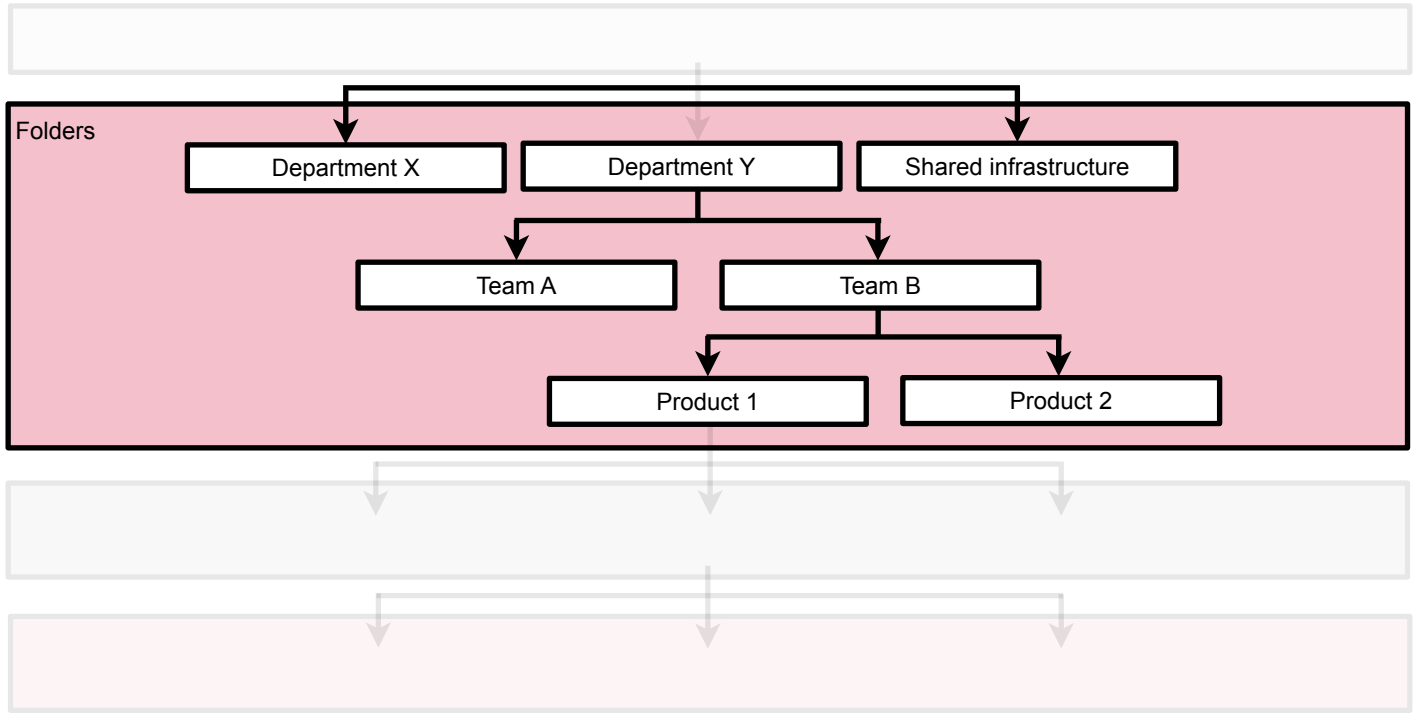


Organization



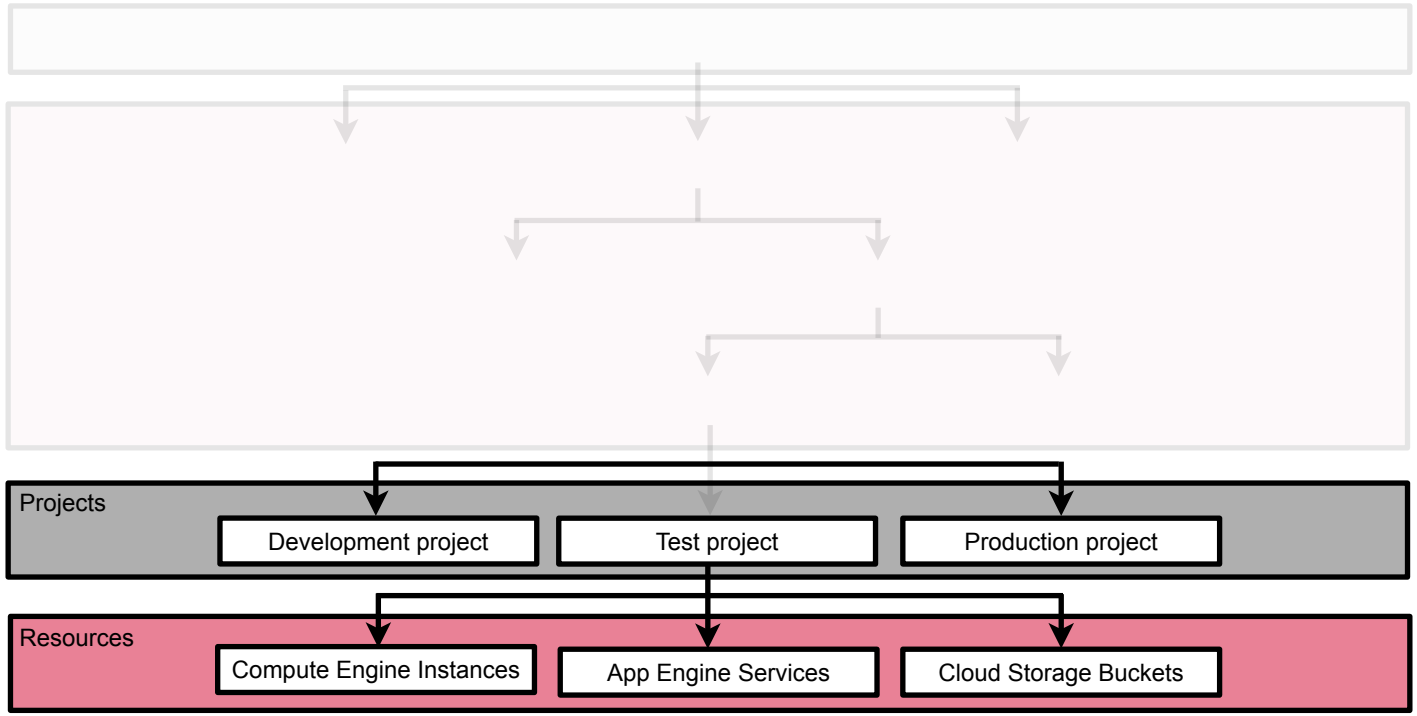
Typically one organization at the root of the hierarchy as a best practice.
Mergers and acquisitions may result in multiple organizations

Folders



Model one folder per department especially if the permissions for department members apply to all projects in a folder

Projects



Environments should always be separated into different projects to be able to manage permissions and resource capacities

Using Google Cloud Resources



Cloud Console

Cloud Shell

Command-line Tools

APIs and Client Libraries

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Hands On Demos - Projects and Cloud Shell



Projects

Which of the following best describes a project on the GCP?

1. Logical grouping of resources based on labels
2. Root node in the resource hierarchy
3. Used to group GCP networks
4. Logical grouping for resources, associated with billing



Projects

Which of the following best describes a project on the GCP?

1. Logical grouping of resources based on labels
2. Root node in the resource hierarchy
3. Used to group GCP networks
4. **Logical grouping for resources, associated with billing**



Cloud Shell

Which of the following best describes Cloud Shell?

1. Command-line utility used to work with the GCP services
2. Ephemeral VM which offers a terminal on the browser
3. PaaS offering on the GCP for hosted applications
4. IaaS offering on the GCP



Cloud Shell

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3. PaaS offering on the GCP for hosted applications
4. IaaS offering on the GCP



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Infra-as-a-Service on the Google Cloud



Choices in Computing





Compute

Where is code executed and how?

Storage

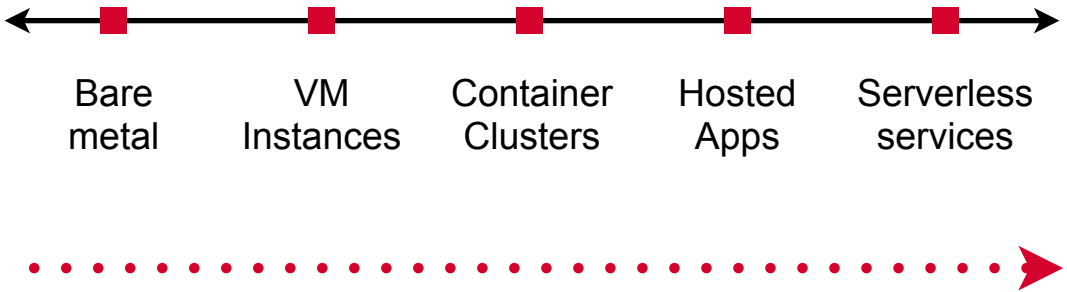
Where is data stored?

Networking, logging, are choices made after this fundamental decision

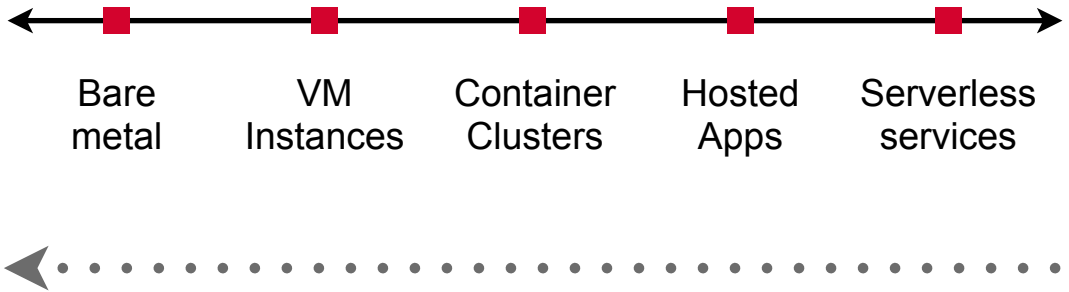
Compute Choices



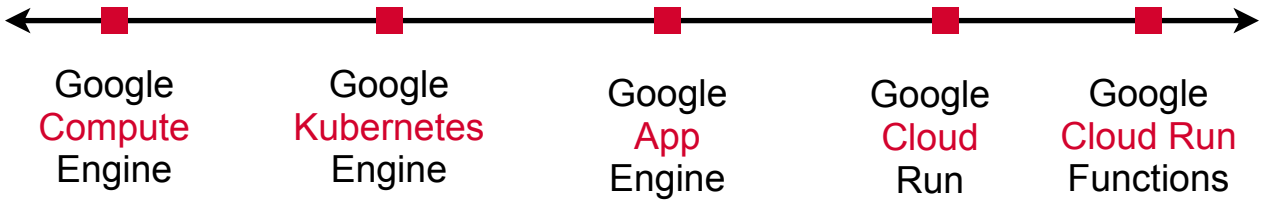
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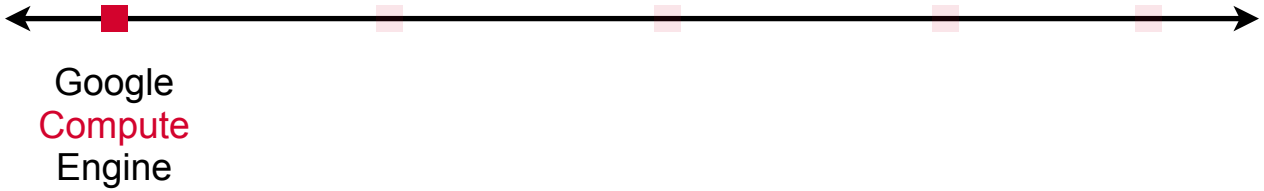
Compute Choices



Google Cloud Compute Choices



Google Cloud Compute Choices



IaaS vs. Bare Metal

Bare Metal

- Apps run on OS which runs on hardware
- Less portable
- CPUs
- Full burden of ops and admin

IaaS

- Hypervisor between apps and hardware
- More portable
- vCPUs
- Much of ops burden managed by service provider

Zones and Regions



Zone

Availability zone (similar to a datacenter)



Region

Set of zones with high-speed network links

Zones and Regions





Zone

“asia-south1-a”

Region

“asia-south1”

Networks are Global Resources



Network

User-controlled IP addresses, subnets and firewalls

Networks are Global Resources



Network

default

Configuration Choices

Machine Family

General purpose, compute optimized, memory optimized, accelerator-optimized

Machine Series

Machines have generation numbers where higher generations have newer features

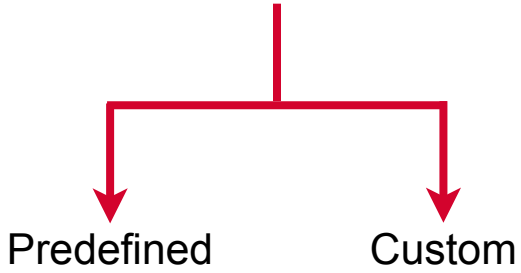
Machine Type

vCPUs count, memory capacity, and storage capacity

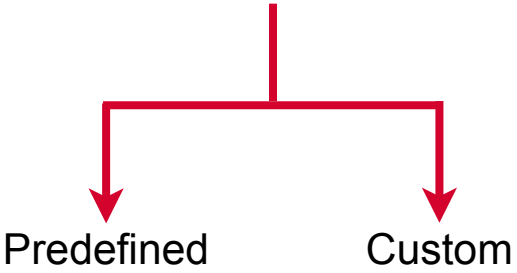
Base Image

Public (free or premium), custom, snapshots from boot disks

Machine Type

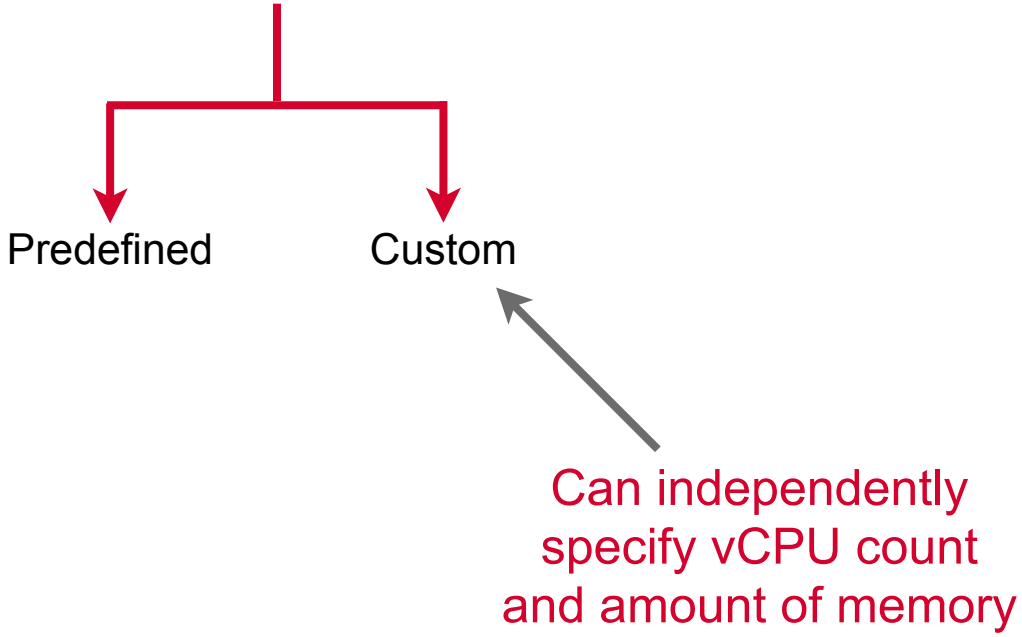


Machine Type

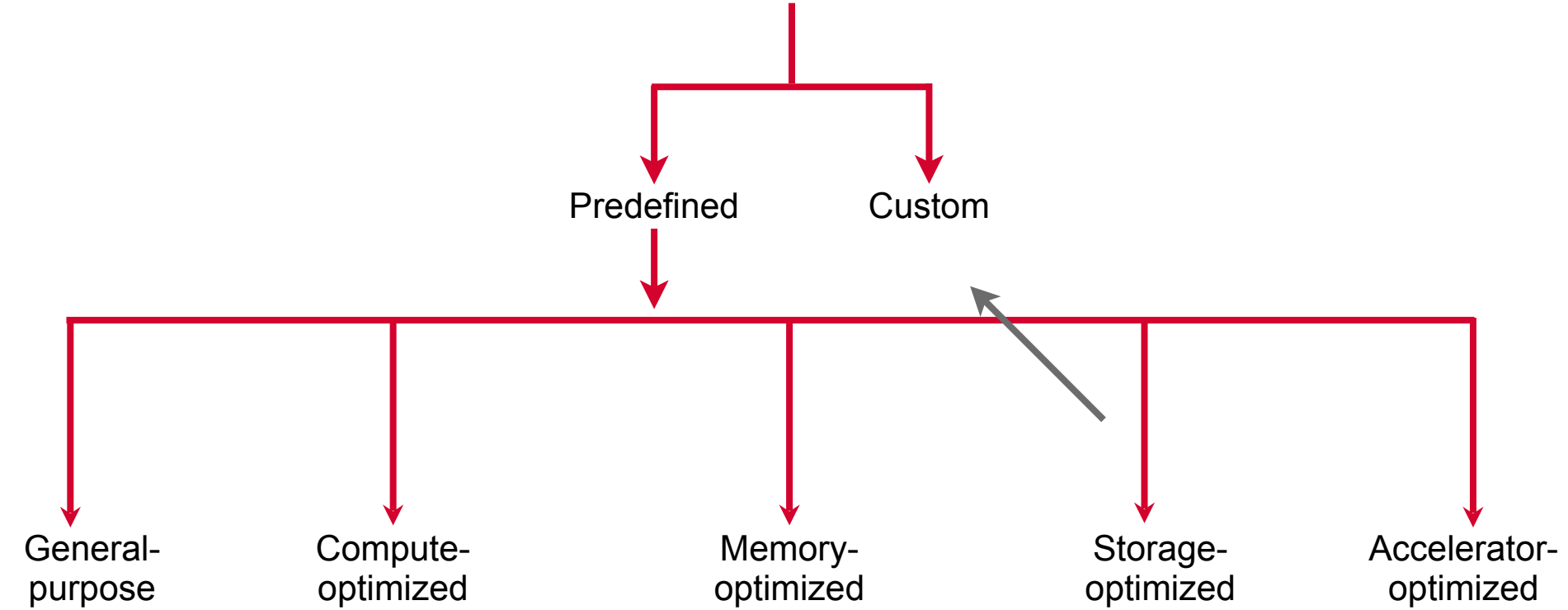


Fixed set of types with fixed ratios of memory to vCPU count

Machine Type



Machine Type

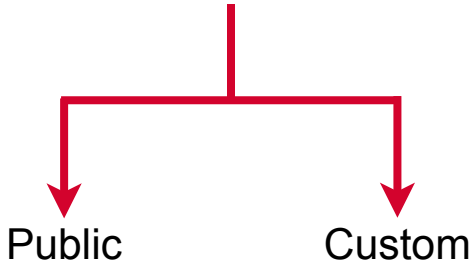


Shared-core Machines

- Cost-effective for running non-resource intensive operations
- A single vCPU run for a time period on single hardware
- Offer micro-bursting capabilities for spikes
- Instance will use additional physical CPUs during spikes



Base Images



Base Images



Provided and maintained by Google, open-source communities, and third-party vendors

All projects have access to these images and can use them to create instances

Base Images



Linux, Windows, Container-optimized OS,
SQL Server

Base Images



Available only to your project

First, create a custom image from boot disks and other images; then, use the custom image to create an instance

Spot VM Instances

An instance that you can create and run at a much lower price than normal instances. However, **GCE might terminate (preempt)** these instances if it requires access to those resources for other tasks.

May not always be available

Not covered by SLAs



Preemptible VM Instances

Similar to Spot VMs (older product and will have fewer features than Spot VMs)

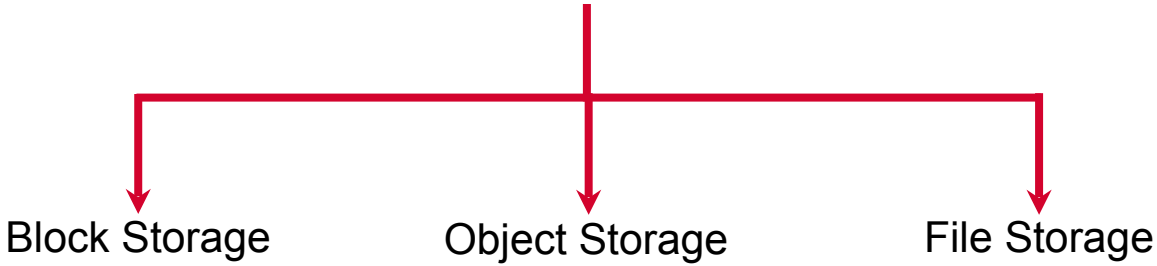
Will definitely be preempted every 24 hours

May not always be available

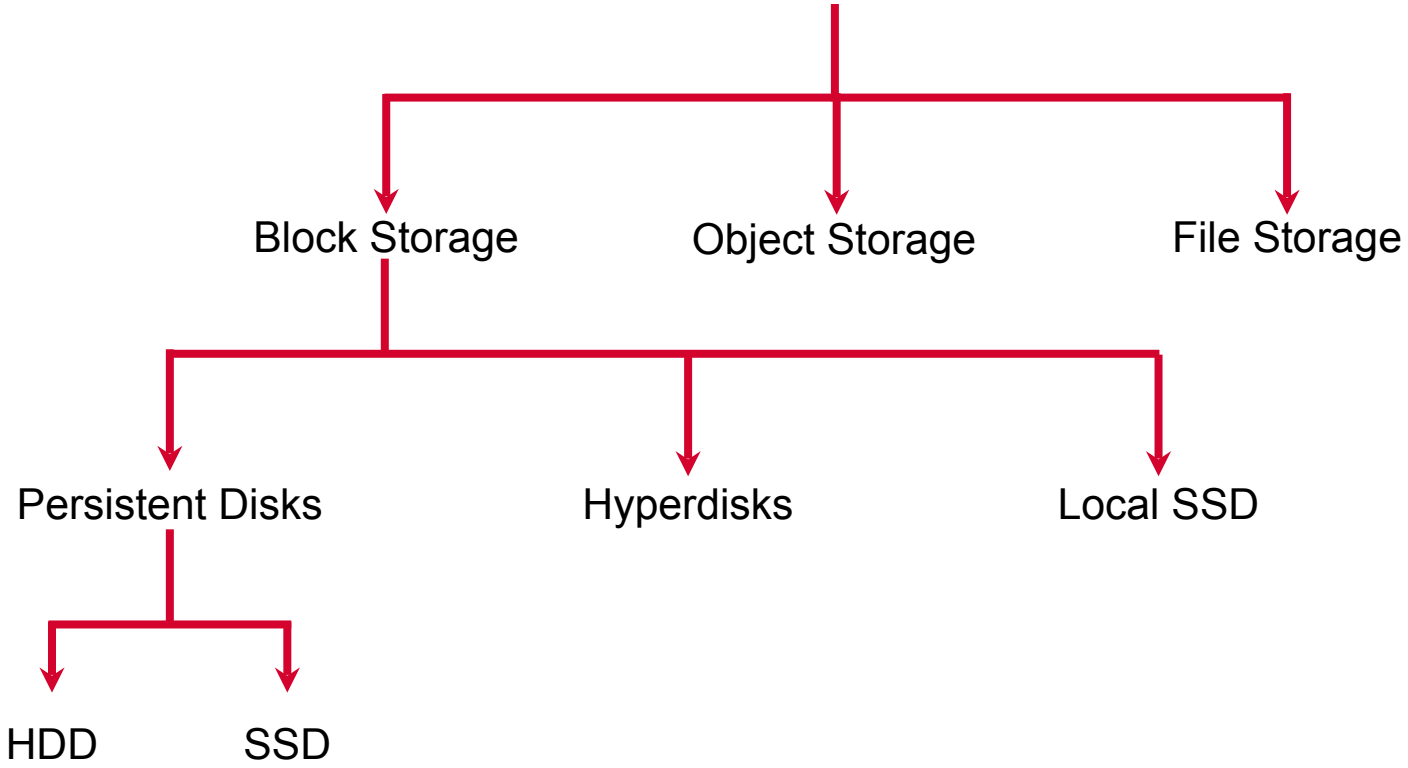
Not covered by SLAs



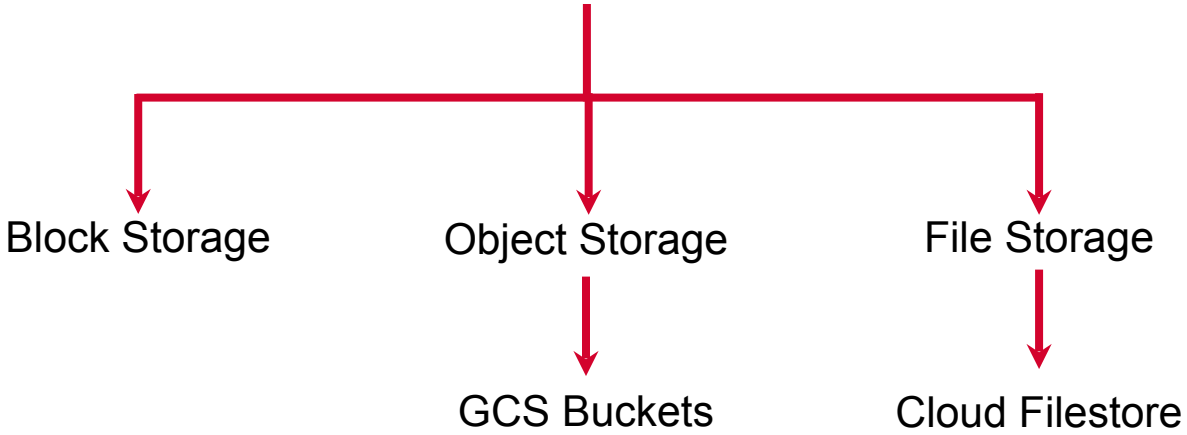
Accessing Storage from VMs



Accessing Storage from VMs



Accessing Storage from VMs



Persistent Disks vs. Buckets

Persistent Disks

- Block storage
- Max 64TB in size
- Pay what you allocate
- Tied to GCE VMs
- Zonal (or regional) access

Buckets

- Object storage
- Infinitely scalable
- Pay what you use
- Independent of GCE VMs
- Global access



Hands On Demos - Google Compute Engine

Region

Which of the following best describes a region on the GCP?

1. A logical area that may be spread across countries
2. A single datacenter on the GCP
3. A geographical area with multiple datacenters
4. Physically connected hardware devices in a datacenter



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Persistent Disks

What is the pricing mechanism for Persistent Disks?

1. If the you create a 100GB disk but you use just 5GB you pay for the entire 100GB
2. If you create a 100GB disk but you use just 5GB you pay for only 5GB
3. If you create a 100GB disk but you use just 5GB you pay for only 5GB + a little extra



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Storage: Exploring Storage Resources on the Google Cloud



Choices in Computing





Compute

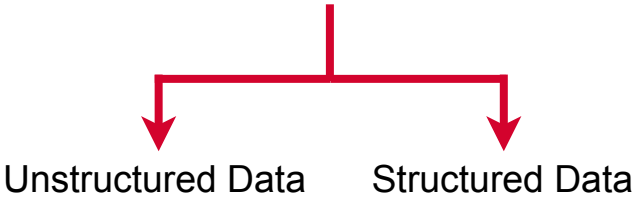
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Storage

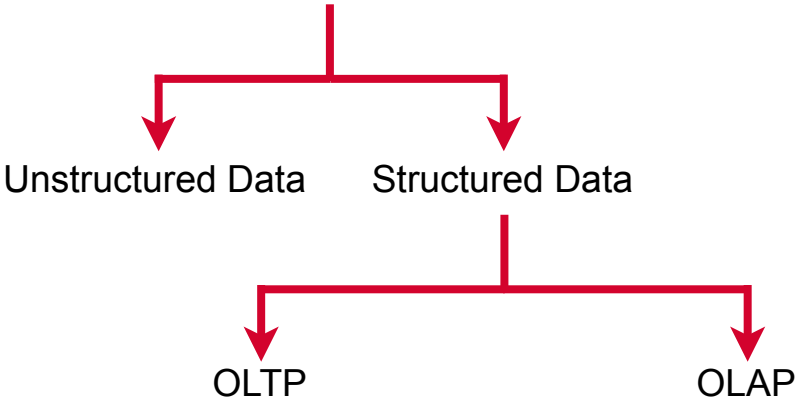
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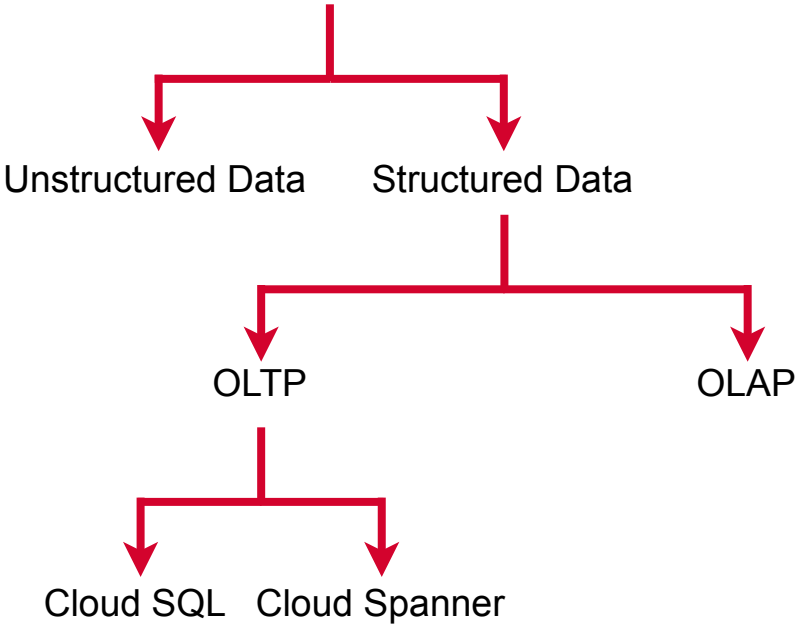
Storage Technologies



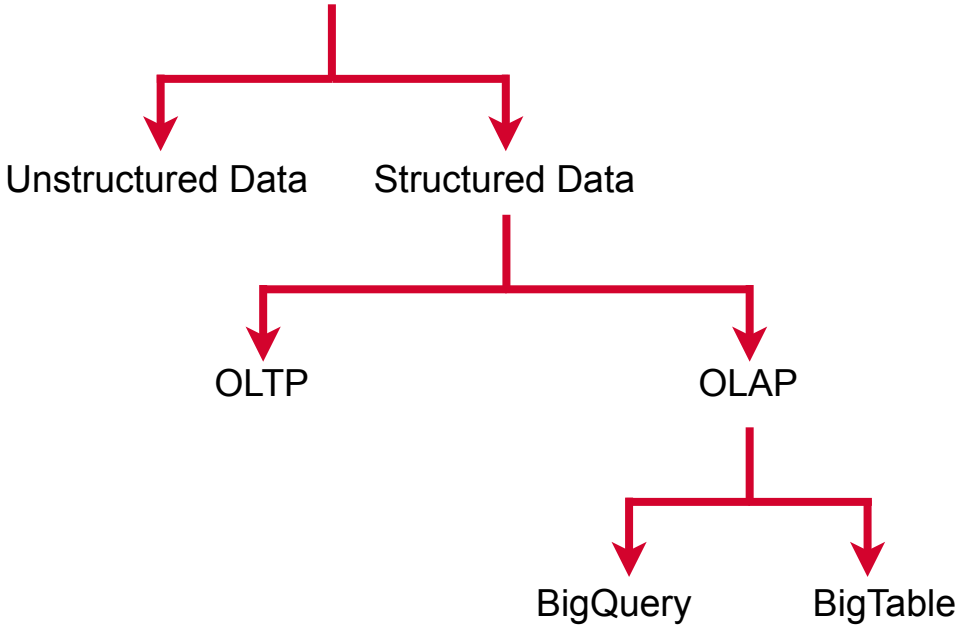
Storage Technologies



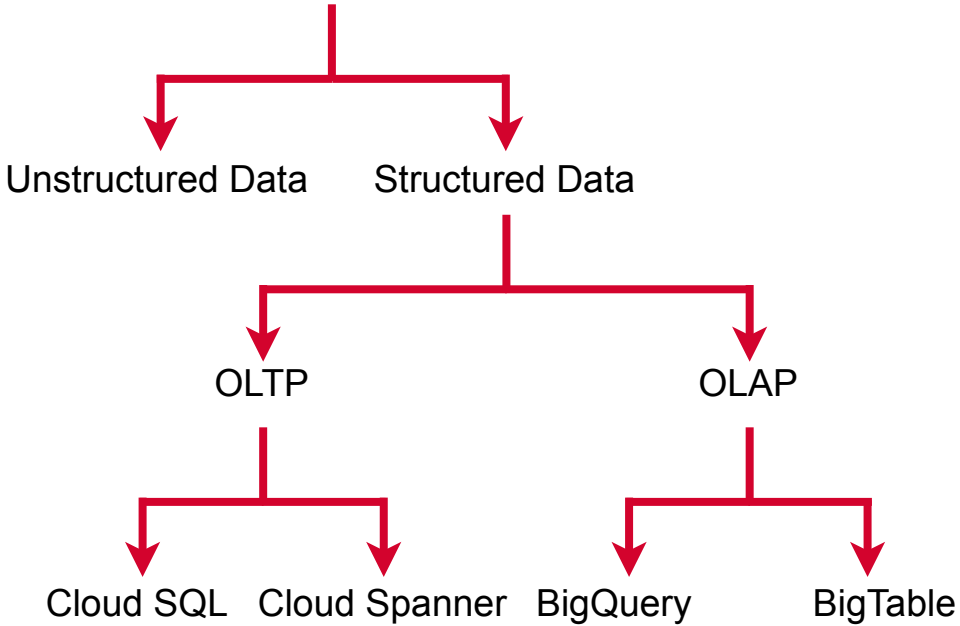
Storage Technologies



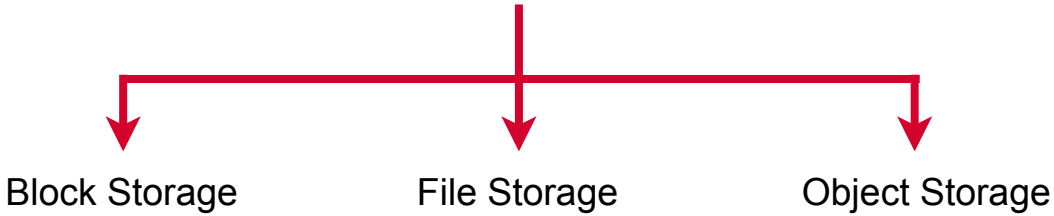
Storage Technologies



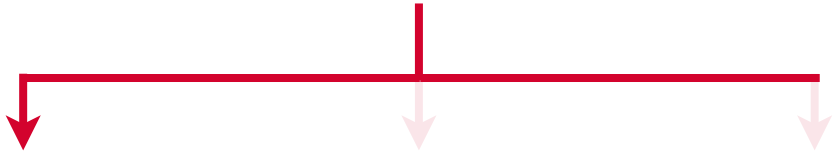
Storage Technologies



Unstructured Data



Unstructured Data



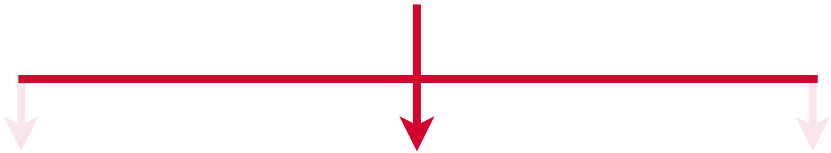
Block Storage

Physically addressable storage accessed from compute - data split into uniform blocks

High performance read and write access at the block level

HDDs/SSDs/
Hyperdisks

Unstructured Data



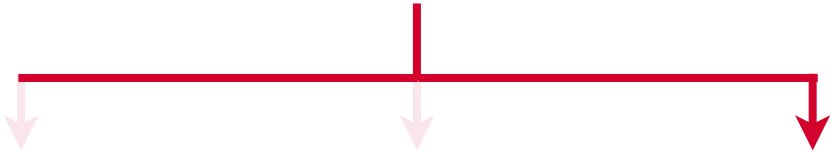
File Storage

Stores data as a hierarchy of files within directories

Shared concurrent access from multiple machines

Cloud Filestore

Unstructured Data



Object Storage

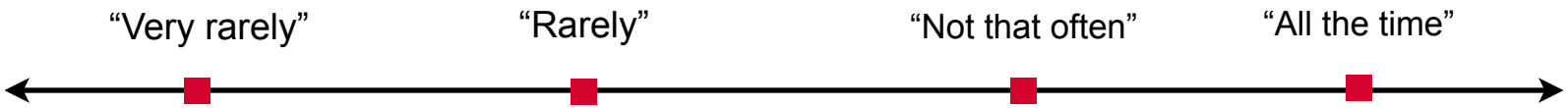
Logically addressable storage accessed from compute or by human users

Cloud Storage

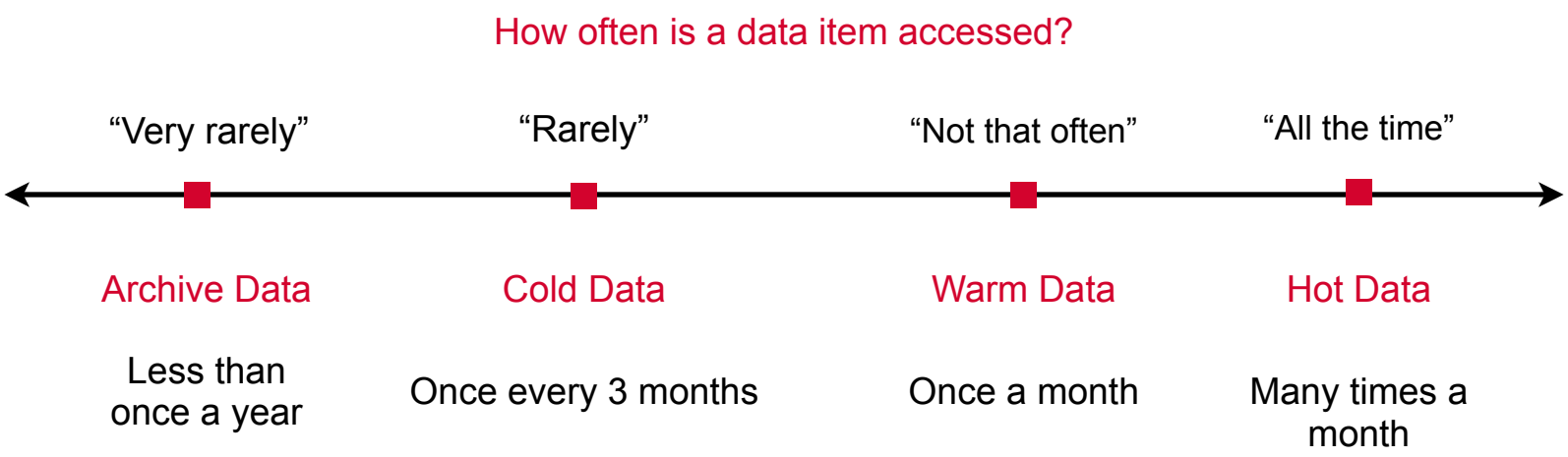

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Cloud
Storage

GCS Storage Classes

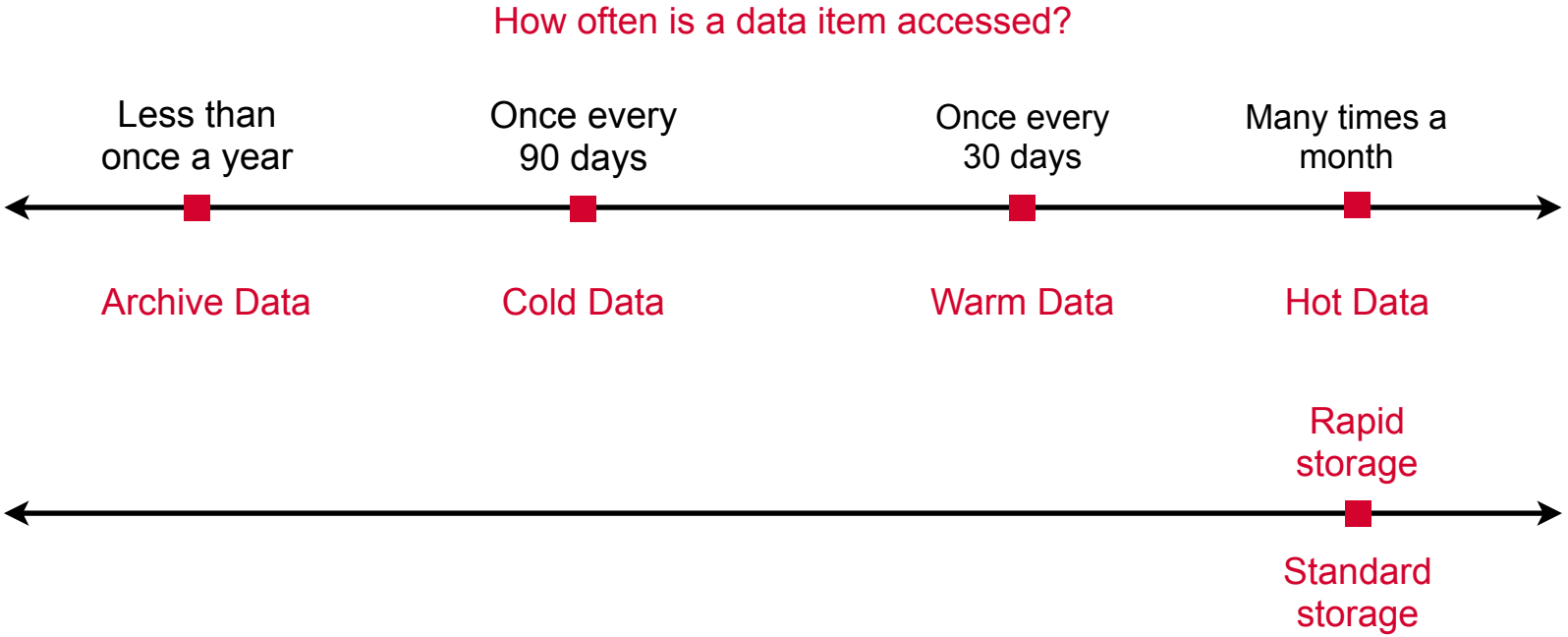
How often is a data item accessed?



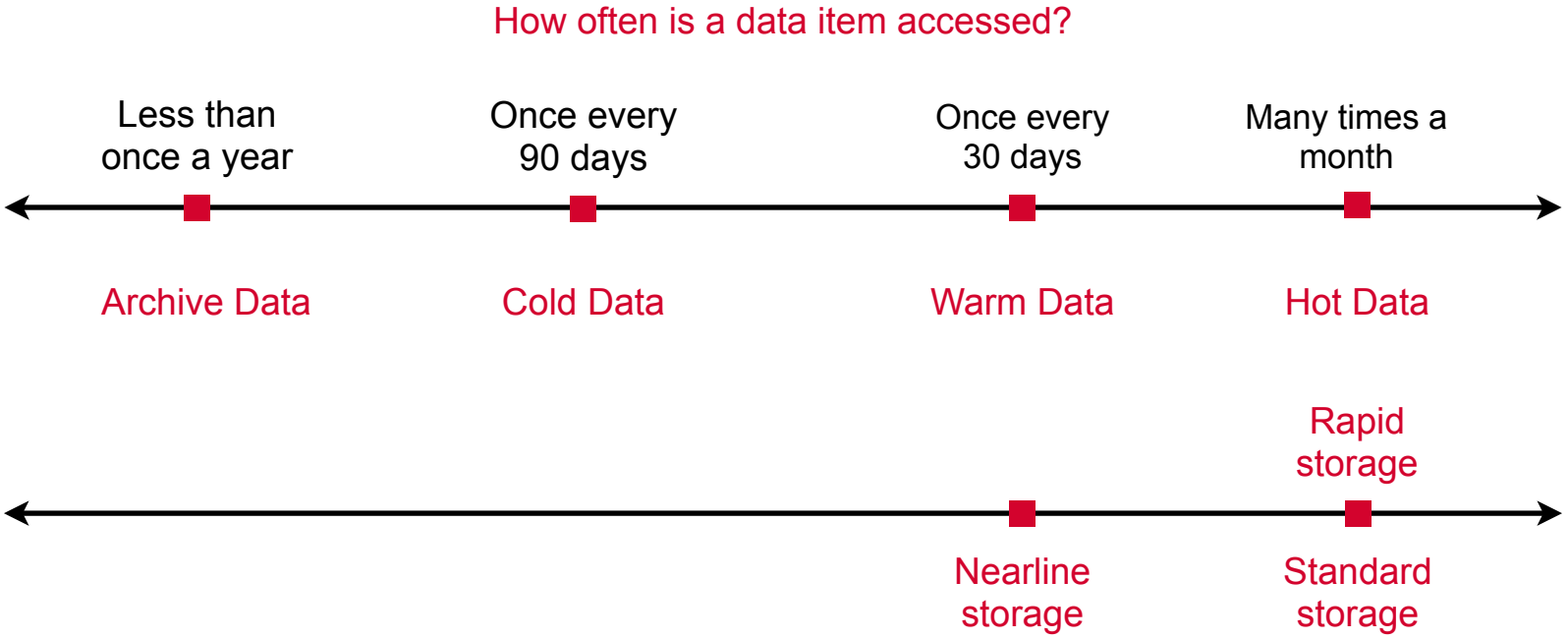
GCS Storage Classes



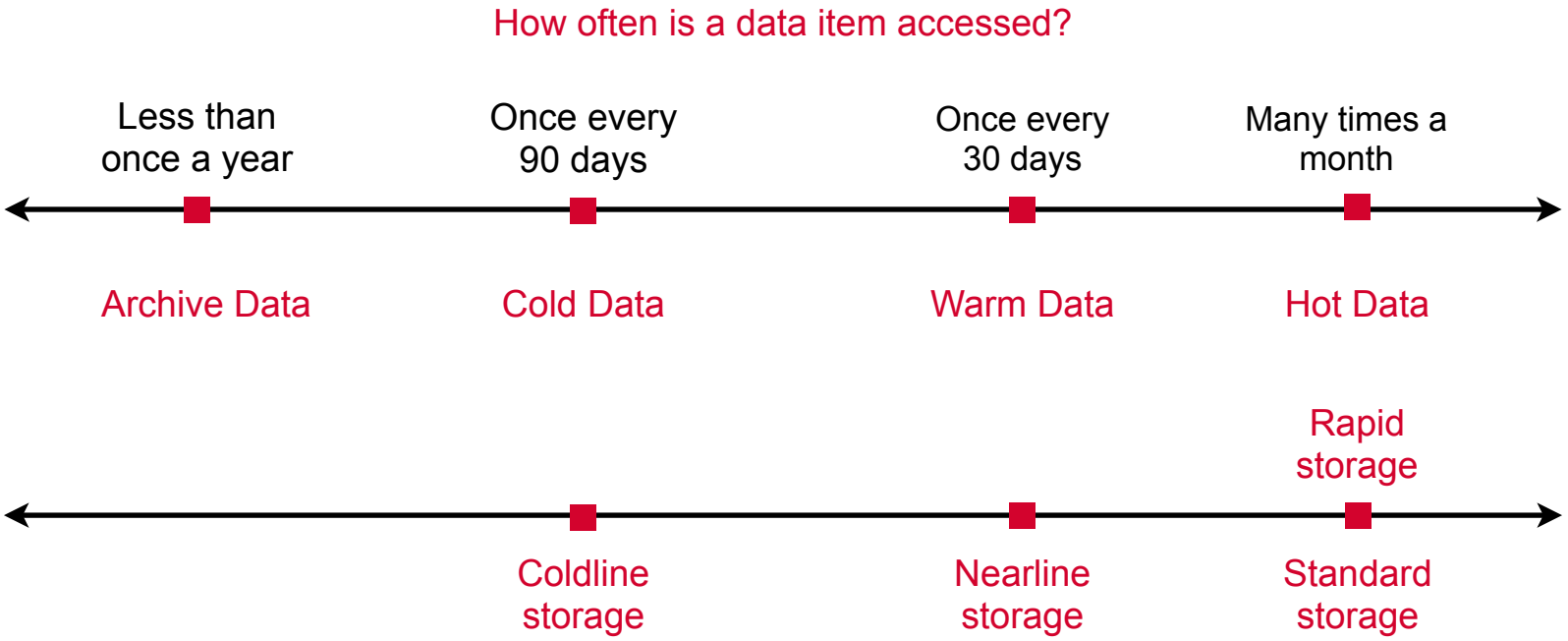
GCS Storage Classes



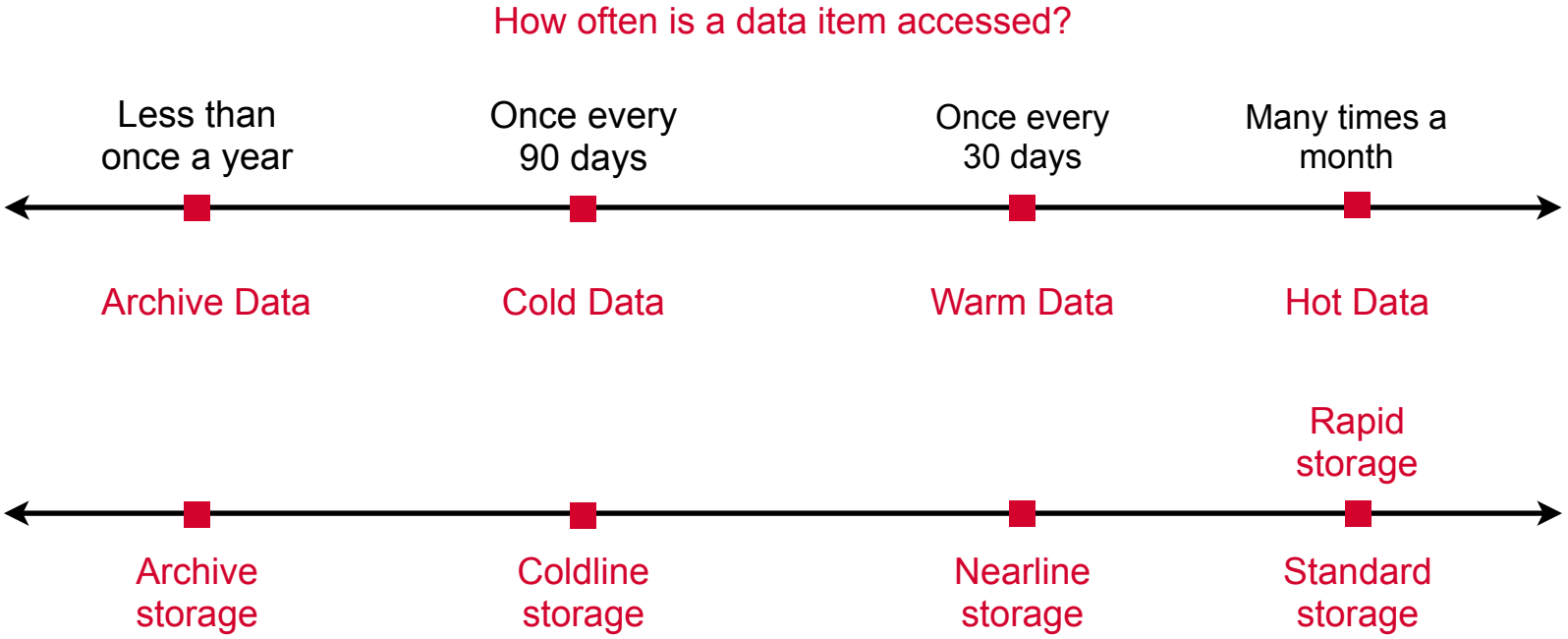
GCS Storage Classes



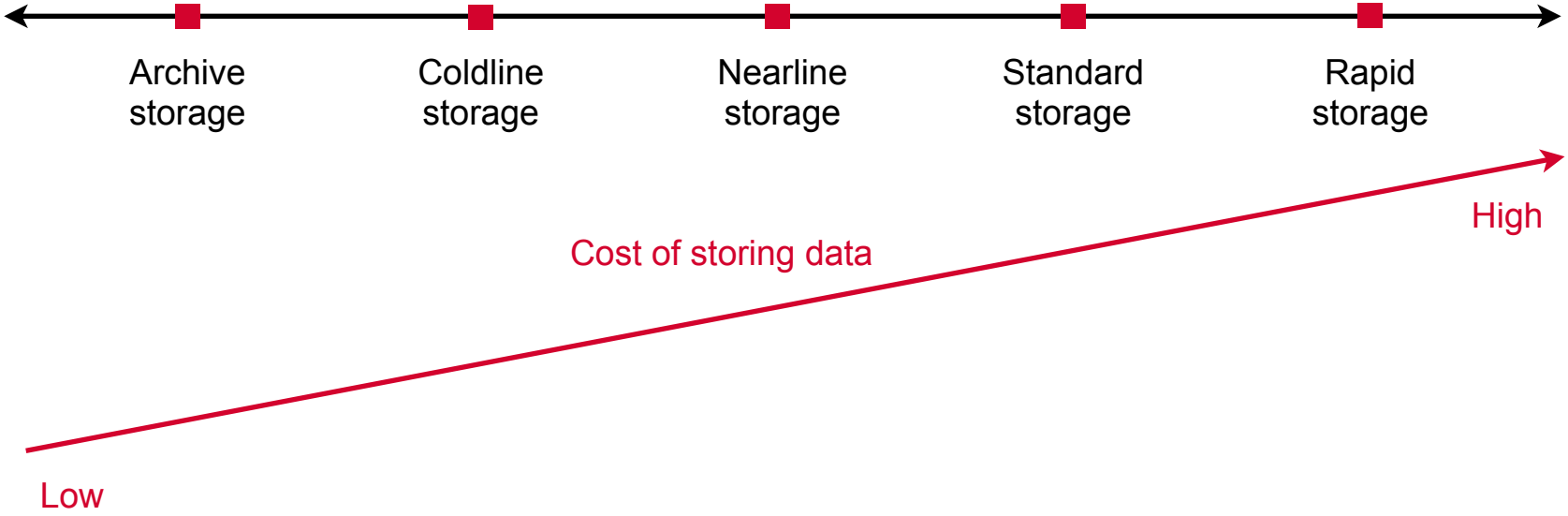
GCS Storage Classes



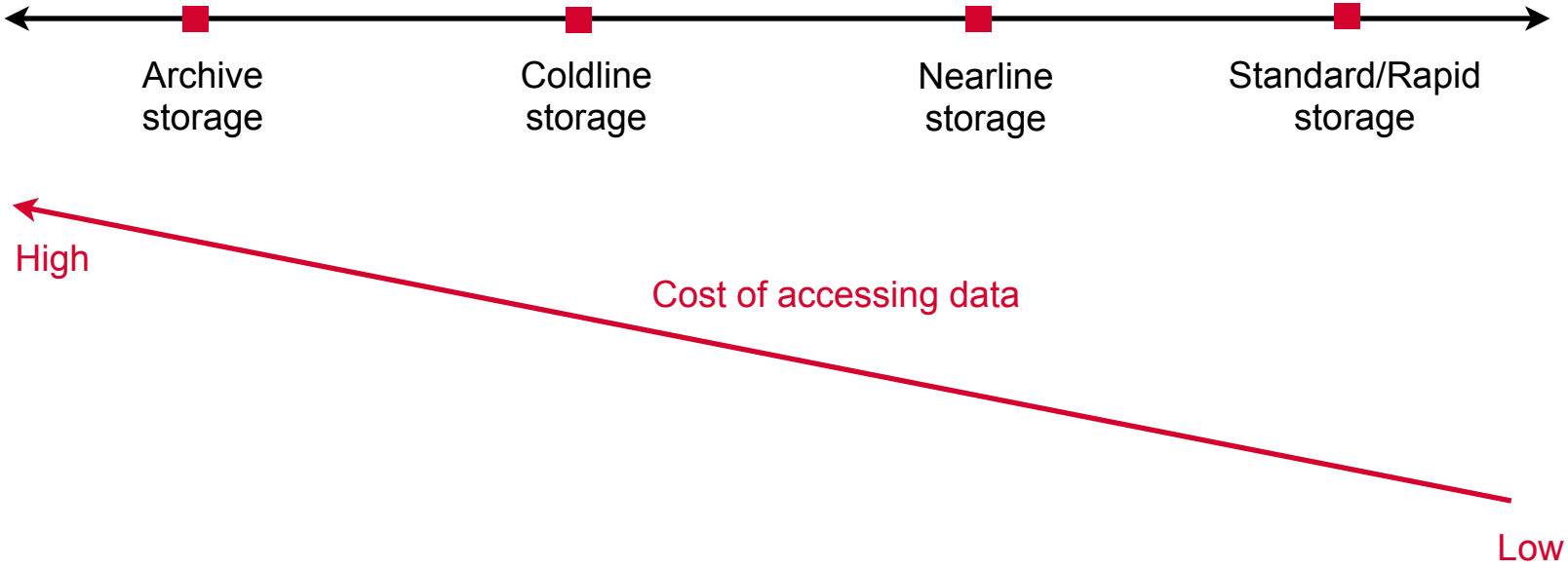
GCS Storage Classes



GCS Storage Classes



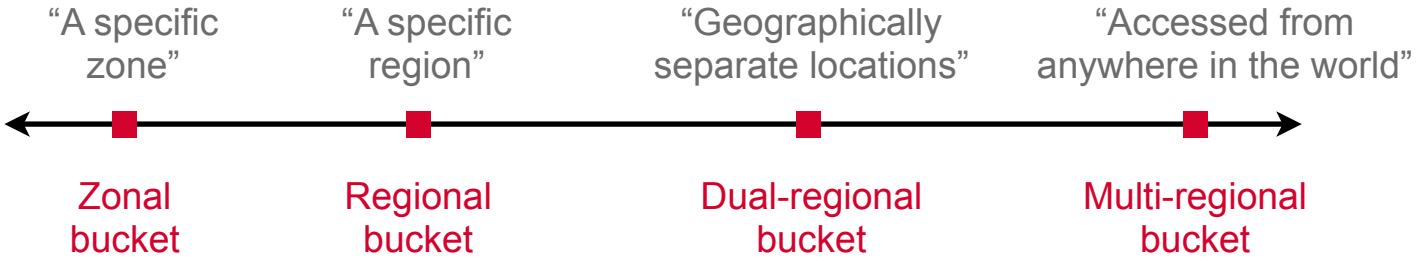
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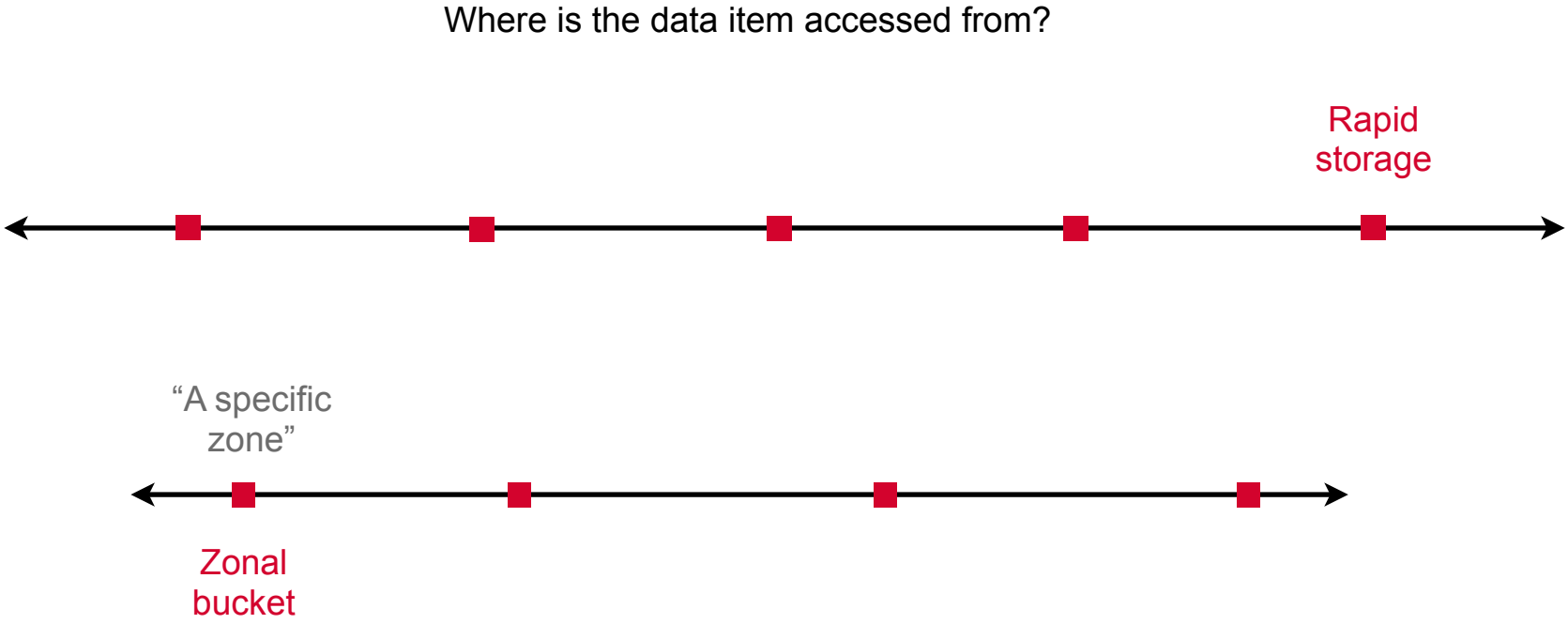
Data Storage Locations

Where is the data item accessed from?





Zonal Buckets Only Possible with Rapid Storage



Rapid Storage

Google Cloud's newest and highest-performance Cloud Storage class, became generally available on April 22, 2026



Rapid Storage

- Storage in a single zone
- Designed for AI/ML and high-performance workloads
- Extremely high throughput (up to ~6 TB/s per bucket)
- Massive scalability (millions of request per bucket)

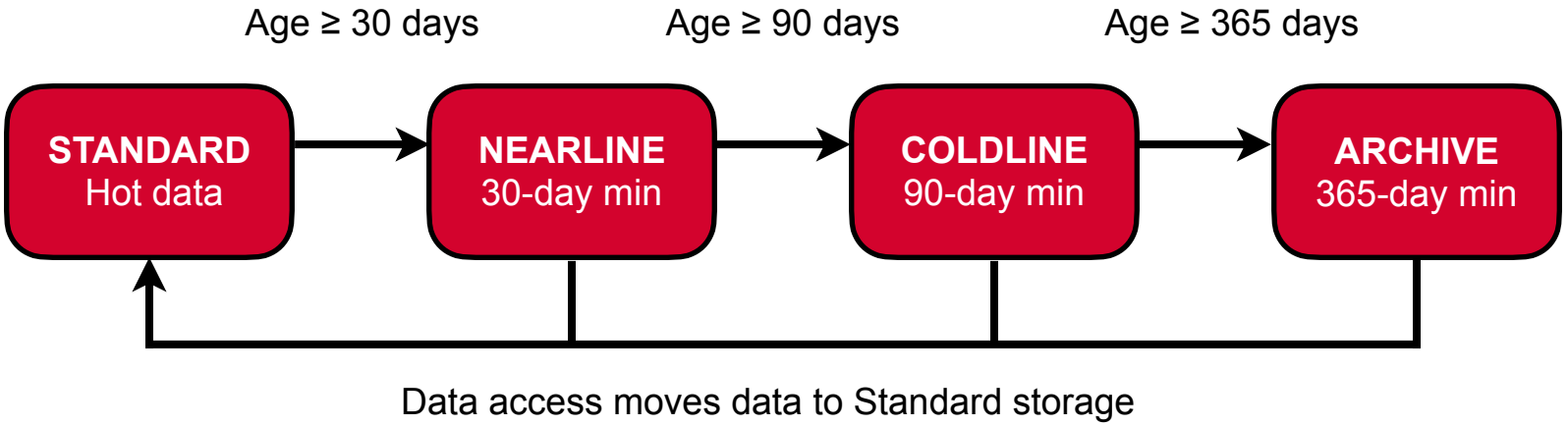


Autoclass

Moves data that is not accessed to colder storage classes to reduce cost

Moves data that is accessed to standard storage to optimize cost of future access

Autoclass


Age ≥ 30 days
Age ≥ 90 days
Age ≥ 365 days
STANDARD
Hot data
NEARLINE
30-day min
COLDLINE
90-day min
ARCHIVE
365-day min
Data access moves data to Standard storage



Coldline and Archive has about the same speed of access as other storage classes (different from AWS Glacier and S3)





- Availability
- Storage Costs
- Retrieval Costs
- Durability
- Access Frequency
- Use Cases

Different storage classes represent different trade-offs

Several parameters along which to compare

Availability

Storage Class	Availability
Rapid storage (zonal)	99.9%
Standard storage (dual and multi-regional)	99.95%
Standard storage (regional)	99.9%
Nearline (regional)	99.0%
Coldline (regional)	99.0%



Dual-region and multi-region buckets are tied to multi-regional locations: US, EU and Asia

Helps adhere to data storage regulations in the US and EU

Storage Costs

Storage Class	Storage Cost (cents/GB/month)
Rapid	11
Standard	2.6
Nearline	1.0
Coldline	0.7
Archive	0.24

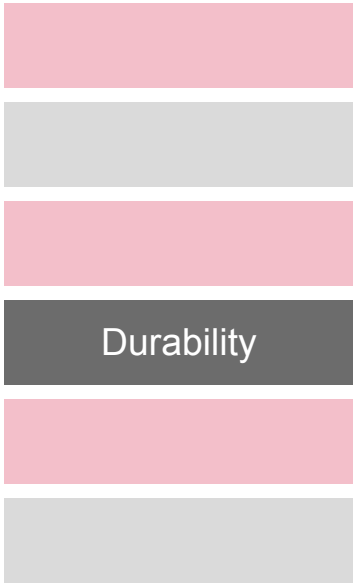
Retrieval Costs

Storage Class	Retrieval Cost (cents/GB)
Rapid	**None**
Standard	**None**
Nearline	1.0
Coldline	2.0
Archive	5.0

Retrieval Costs

Storage Class	Minimum Commitment
Rapid	**None**
Standard	**None**
Nearline	30 days*
Coldline	90 days*
Archive	365 days*

*Early deletion will incur charges



Storage Class	Durability
Rapid	99.999999999%
Standard	99.999999999%
Nearline	99.999999999%
Coldline	99.999999999%
Archive	99.999999999%

“11 nines”

Access Frequency

Storage Class	Access Frequency
Rapid	Daily
Standard	Daily
Nearline	Monthly
Coldline	Quarterly
Archive	Less than once a year

Use Cases

Storage Class	Access Frequency
Rapid storage	AI/ML training workloads
Standard storage (dual and multi-regional)	Serving websites, interactive workloads, mobile and gaming applications
Standard storage (regional)	Access from Compute Engine VMs or Dataproc cluster
Nearline	Data backup, disaster recovery, archival storage
Coldline/Archive	Legal or regulatory needs; also disaster recovery where recovery time is important

Autoclass

Moves data that is not accessed to colder storage classes to reduce cost

Moves data that is accessed to standard storage to optimize cost of future access

Soft Delete

- Default 7 day retention for deleted objects
- Prevents accidental deletion



Object Versioning

- Needs to be enabled for bucket
- Once enabled, bucket creates archived versions of each object
- Whenever live object is overwritten or deleted
- Version with unique generation number is created
- Each copy charged separately



Object Lifecycle Management

- Can automatically specify changes to object storage class
  - “Change from standard to nearline after 30 days”
  - “Delete all data created before 1/8/2024”
  - “Delete all but 2 most recent versions”



Encryption

- Encrypted even at rest
- Default: Google generates keys
- Can use CSEK
  - *C*ustomer *S*upplied *E*ncryption *K*ey




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Hands on Demos - Google Cloud Storage

Storage Class

Which of the following is true for coldline storage?

1. Low cost of storage, high cost of retrieval
2. Low cost of storage, low cost of retrieval
3. High cost of storage, low cost of retrieval
4. High cost of storage, high cost of retrieval



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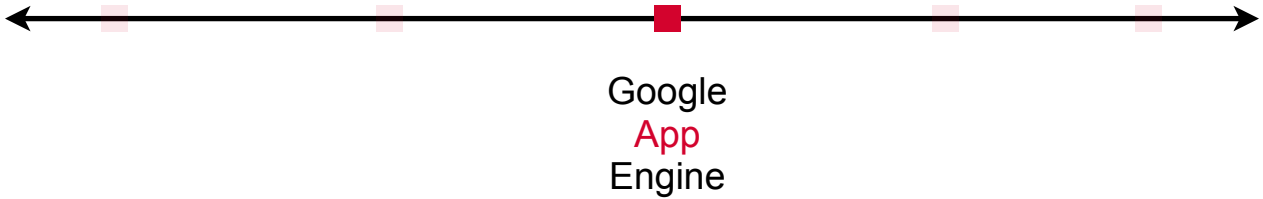
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Platform-as-a-Service on the Google Cloud




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Google
App
Engine

Google Cloud Compute Choices



Google App Engine



Web framework and platform for hosting web applications on the Google Cloud Platform

Support for Go, PHP, Java, Python, Node.js, .NET, Ruby and other languages

Google App Engine



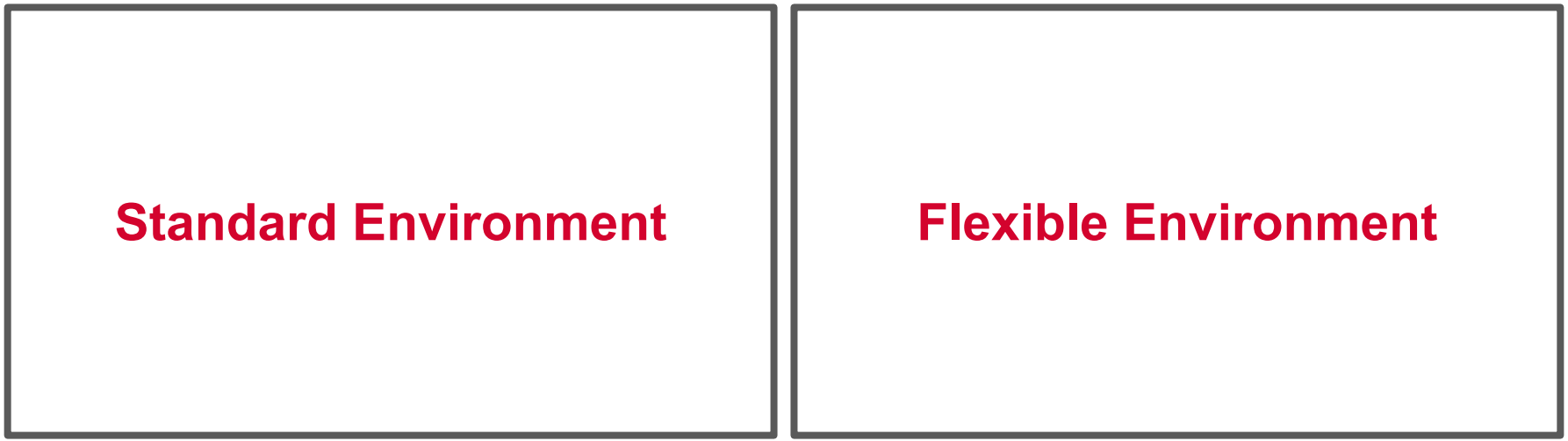
Web framework and platform for hosting web applications on the Google Cloud Platform

Support for Go, PHP, Java, Python, Node.js, .NET, Ruby and other languages

Focus on development and code

Infrastructure and scaling taken care of by the platform

App Engine Environments



App Engine Environments

Standard

- App runs in a proprietary sandbox
- Instances start up in seconds
- Code in few languages/versions only
- No other runtimes possible
- Apps cannot access Compute Engine resources
- Can install 3rd party binaries for selected runtimes

App Engine Environments



Standard

- App runs in a proprietary sandbox
- Instances start up in seconds
- Code in few languages/versions only
- No other runtimes possible
- Apps cannot access Compute Engine resources
- Can install 3rd party binaries for selected runtimes

Flexible

- Runs in Docker container on GCE VM
- Instance start up in minutes
- Code in far more languages/versions
- Custom runtimes possible
- Apps can access Compute Engine resources, some OS packages
- Can install and access third-party binaries

App Engine Environments



Standard

- Apps that experience traffic spikes
- Usually stateless HTTP web apps

Flexible

- Apps that experience consistent traffic
- General purpose apps


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Google
Cloud Run
Functions

Google Cloud Compute Choices



Cloud Run Functions



Event-driven serverless compute platform

Event-driven Serverless Compute



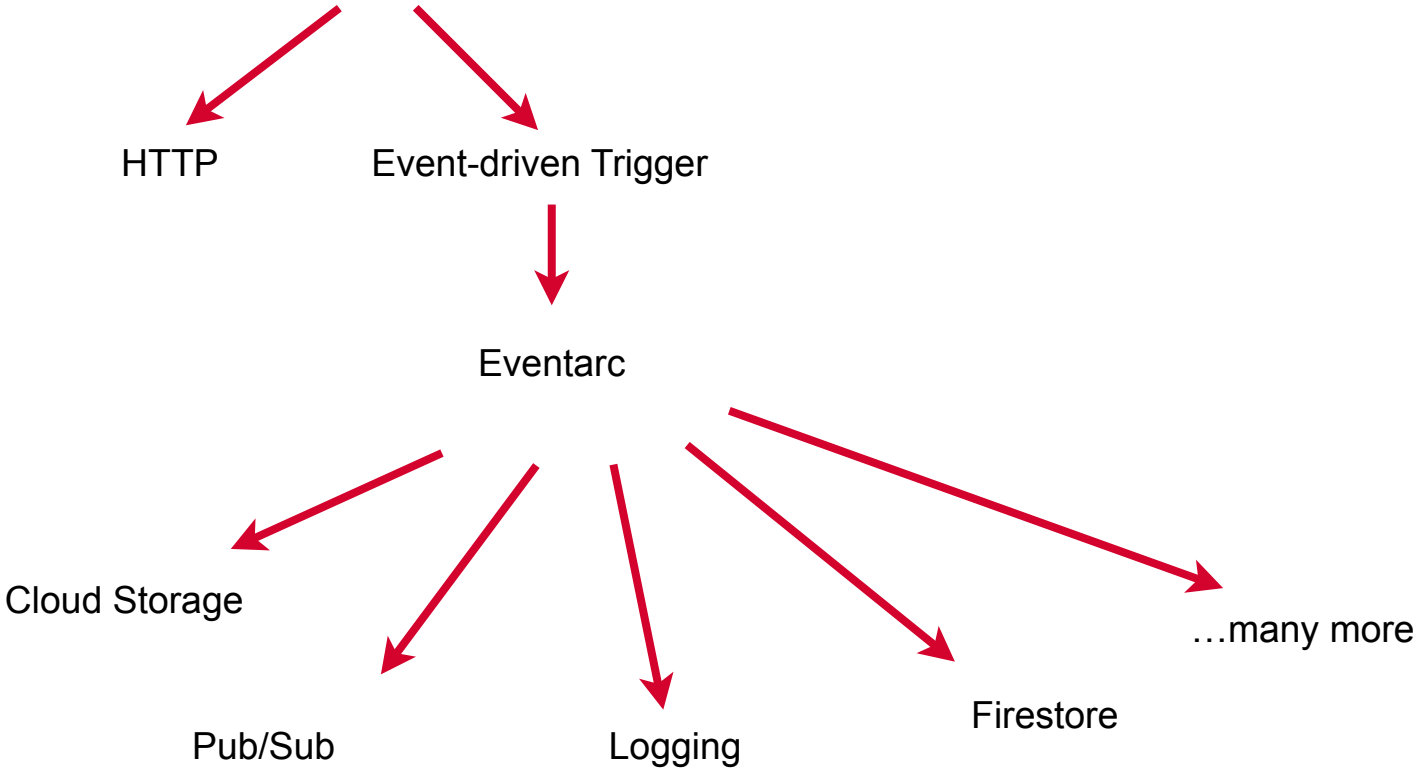
Event occurs

Platform triggers execution

Cloud Run Function code runs

Invokes other Google Cloud services

Types of Events



Eventarc

- Event routing service - delivers events from Google and custom sources to Google services destinations
- Destinations include
  - Cloud Run Functions
  - Cloud Run
  - Workflows
  - GKE services
  - Others
- Unifies the trigger mechanism for every service



Concurrency and Scale

- Spin up function instances based on current load
- Functions receive event parameters from platform
- Functions do not share memory or variables
- An instance processes a single request (generation 1)
- Function concurrency supported (generation 2)
- Functions should be stateless



Google Cloud Compute Choices



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Google Cloud Run



Cloud Run



Serverless, managed platform that lets you run containers directly on top of Google’s scalable architecture

Cloud Run

- Write your code in any programming language
- Create a container image (or use source-based deployment option - Google Cloud will build container image for you)
- Register the container with the artifact registry
- Deploy your container directly using Cloud Run
- No cluster creation no infrastructure management
- Request-based pricing and instance-based pricing



Running Code Using Cloud Run



Both use the same environment and have the same integrations with other Google Cloud services

Cloud Run Services

- Used to run code that responds to web requests or events
- Each service located in a Google Cloud region
- Replicated across zones in the region
- Exposes an endpoint
- Automatically scales underlying infrastructure to handle incoming requests
- Version management, rollbacks, traffic management - all handled by the platform

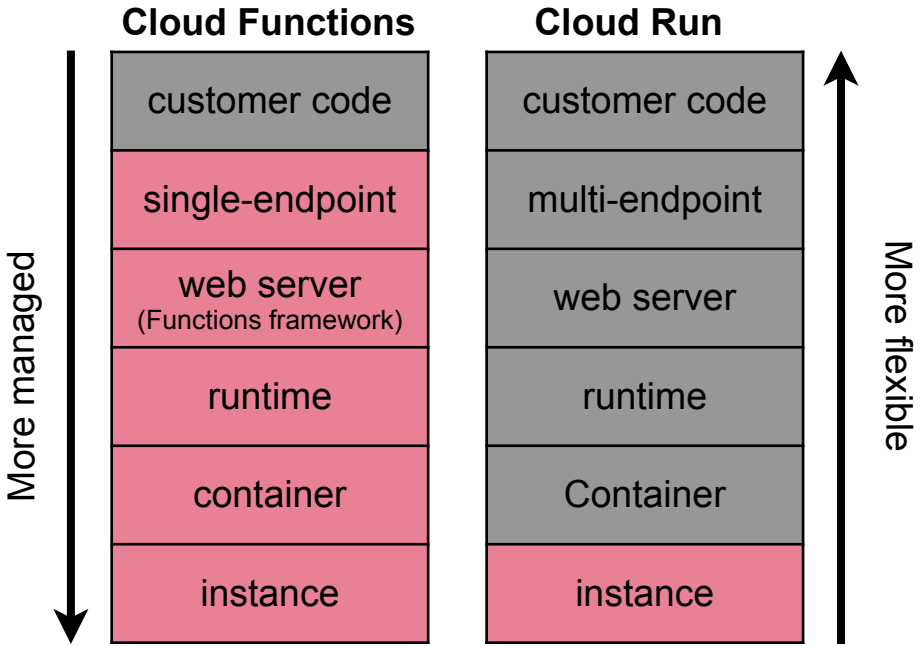


Cloud Run Jobs

- Used to run code that performs work (a job) and quits when the job is done
- Each service located in a Google Cloud region and executes one or more containers to completion
- A job comprises of many tasks executing in parallel - each container runs one task



Cloud Functions vs. Cloud Run



How managed do you want to be?

Cloud Functions vs. Cloud Run



Cloud Functions

- Specific limited runtimes supported
- Can be triggered based on platform events
- No support for running jobs
- 2nd generation functions support concurrency

Cloud Run

- All runtimes that can be run using containers
- Expose endpoints and invoked using HTTP requests
- Support for running jobs
- Great support for concurrent requests

Cloud Functions vs. Cloud Run

Cloud Functions

- Choose Cloud Functions if you primarily want to connect to other cloud services on Google Cloud

Cloud Run

- Choose Cloud Run if you want a simple way to scale and maintain services using containers

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Hands on Demos - Google Cloud Functions



Serverless Applications

When would you choose to use Cloud Functions over Cloud Run?

1. When you need to run a containerized application.
2. When you need to run a function in response to events.
3. When you require fine-grained control over application resources.
4. When you need to deploy a long-running application.



Serverless Applications

When would you choose to use Cloud Functions over Cloud Run?

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Networking on the Google Cloud



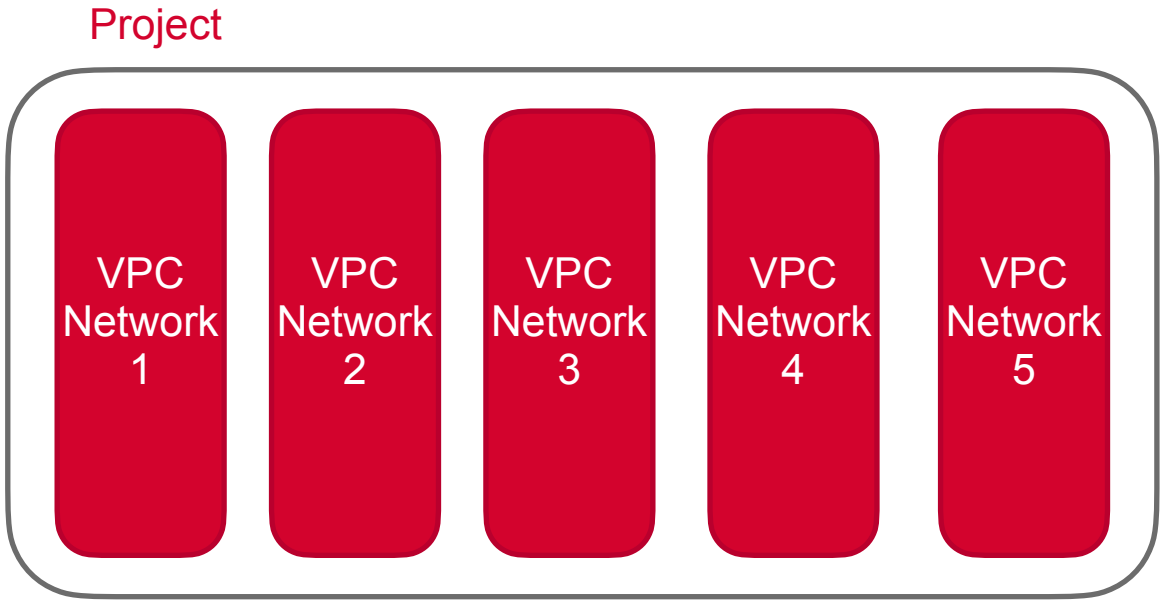
Google Virtual Private Cloud

A VPC network, or just network, is a global, private, isolated virtual network partition that provides managed network functionality on the Google Cloud

Google Virtual Private Cloud

A VPC network, often just called a network, is a global, private, isolated virtual network partition that provides managed network functionality on Google Cloud

Multiple VPCs in a Project

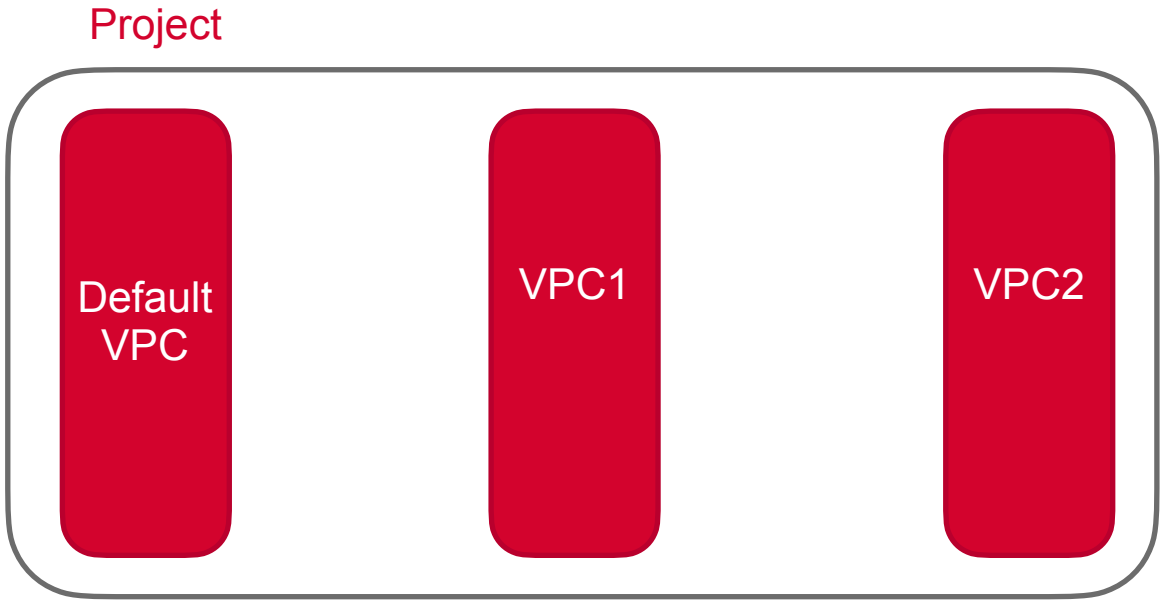


Projects and VPCs

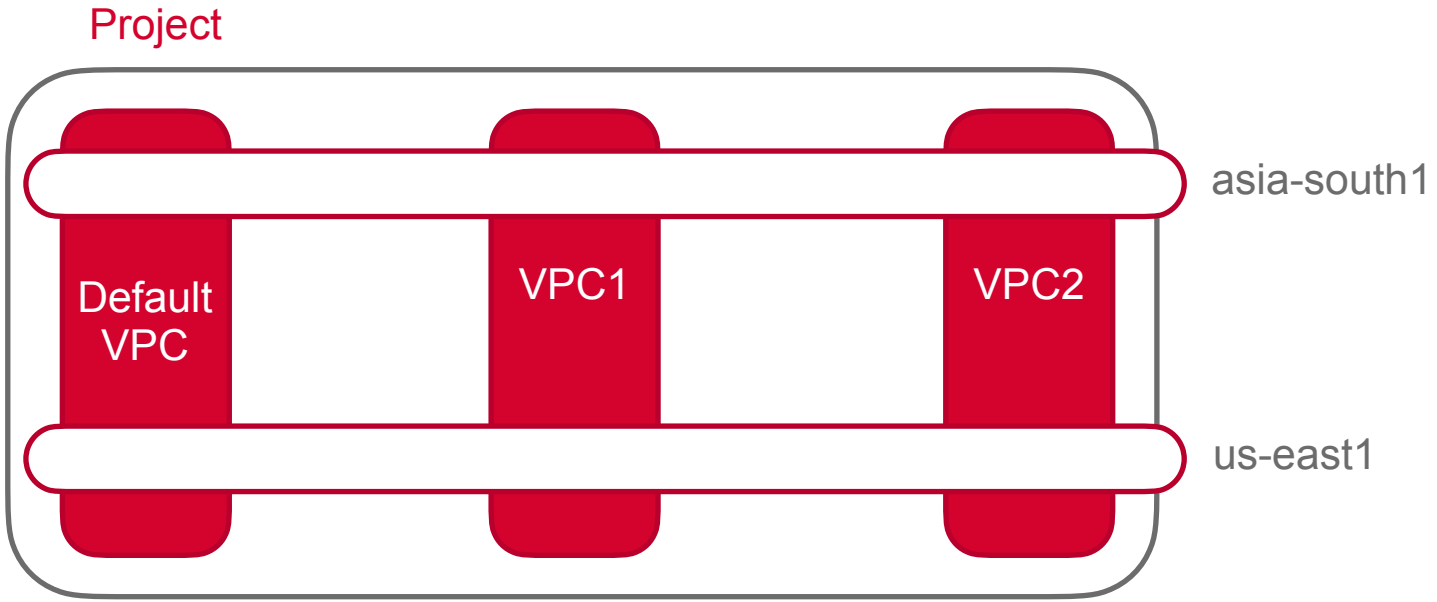
- VPCs are global resources on Google Cloud
- Each VPC must exist inside a project
- Default VPC pre-created in each project
- Can add additional VPCs
  - Auto Mode
  - Custom Mode



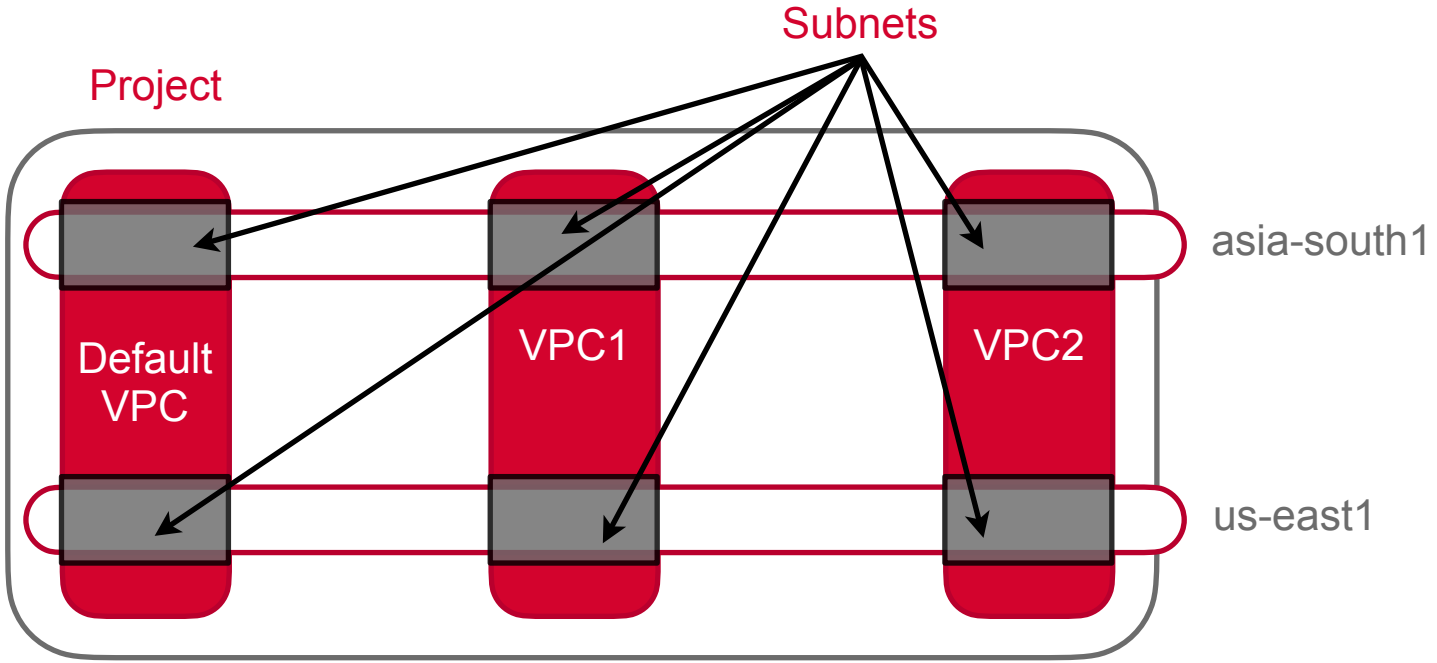
VPCs Are Global



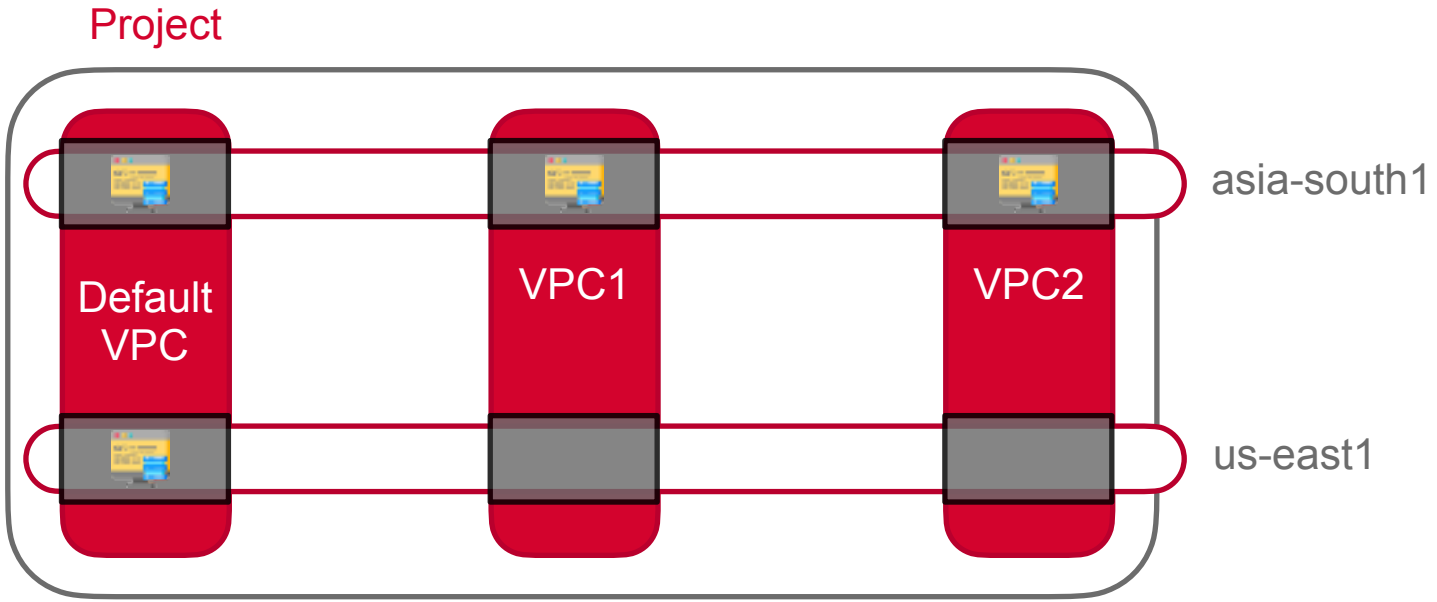
VPCs Are Global



Subnets in Each Region



Resources Provisioned on Subnets

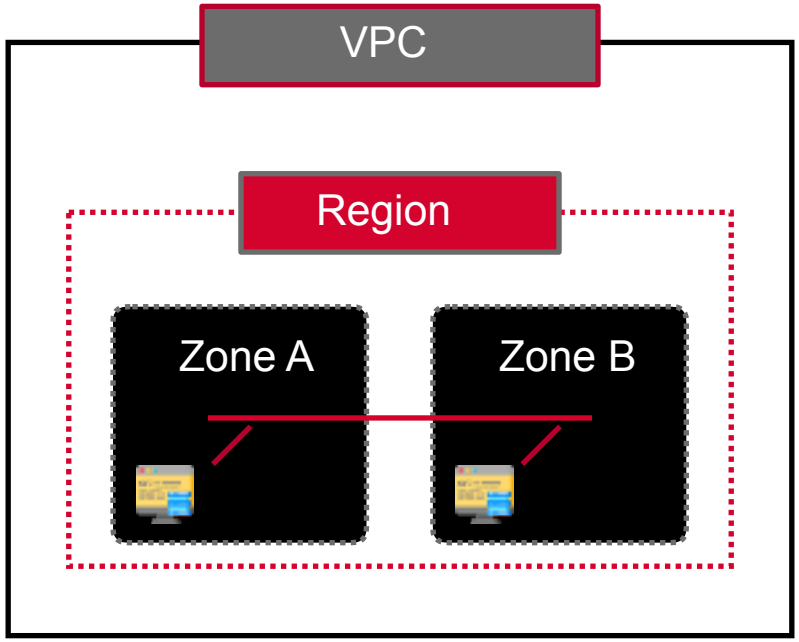


Subnets

- **IP range partitions** within global VPCs
- VPCs have no IP ranges
- Subnets are regional - can span zones inside a region
- Network has to have at least one subnet before you can use it



Subnets Span Zones



Subnets and IP Ranges

- Each subnet must have primary address range
- Valid RFC 1918 CIDR block
- Subnet ranges in **same network cannot overlap**
- Subnet ranges in **different networks can overlap**



AutoMode and CustomMode VPCs

Auto Mode

Subnets automatically created in each region, default firewall rules

Custom Mode

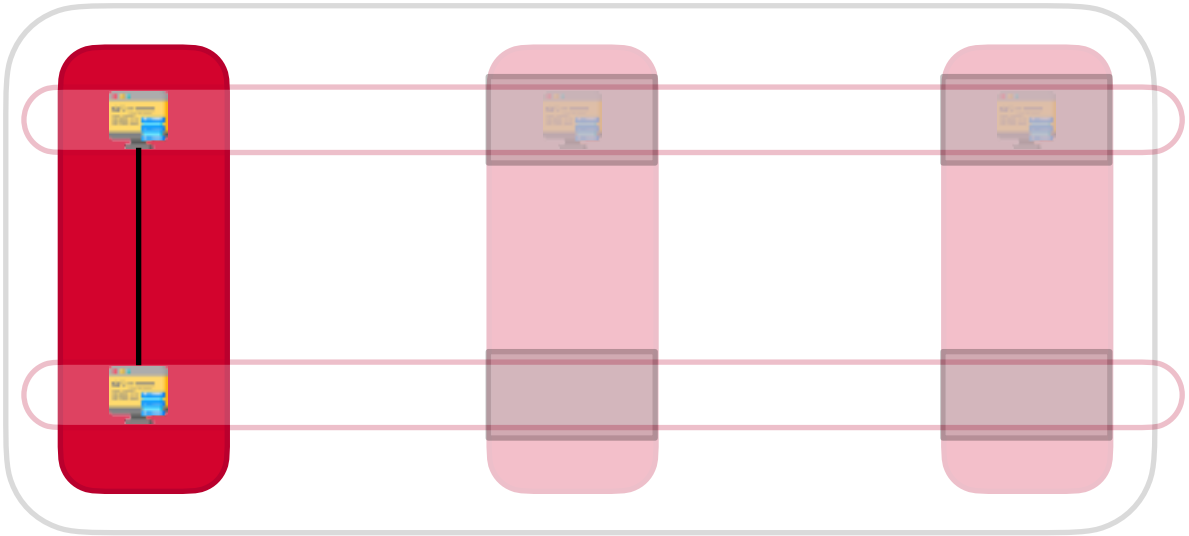
Manually create subnets in regions, no defaults preconfigured

AutoMode and CustomMode VPCs

- Auto Mode VPCs have pre-created subnets
  - One in each Google Cloud region
- Custom Mode VPCs start with no subnets
  - Full control over which regions have subnets
  - Can create multiple subnets in a region

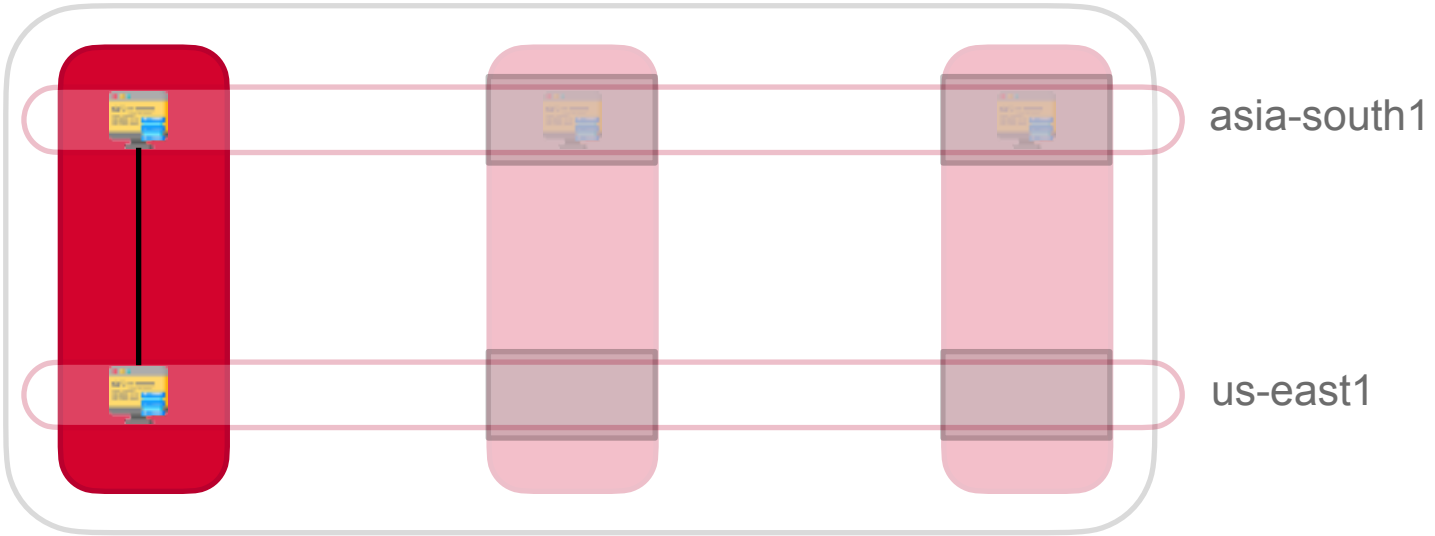


Communication on VPCs



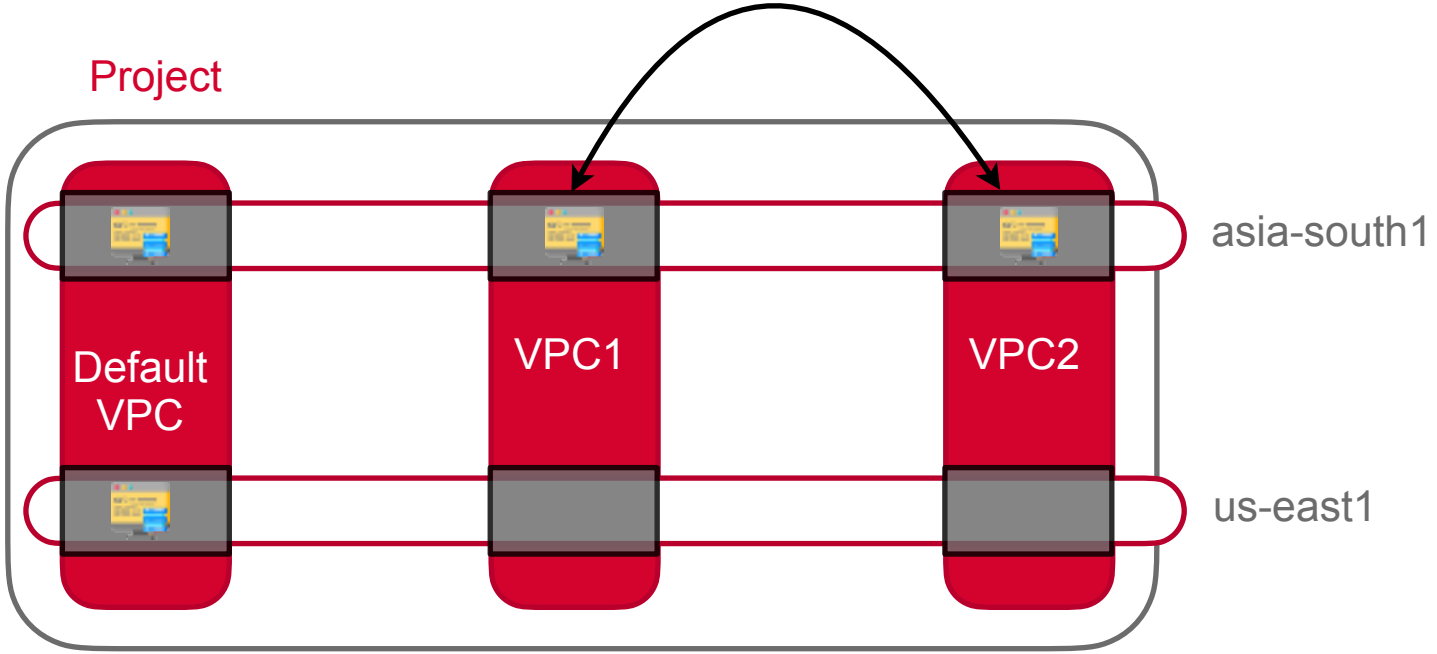
Resources within a VPC communicate using
private IP addresses

Communication on VPCs



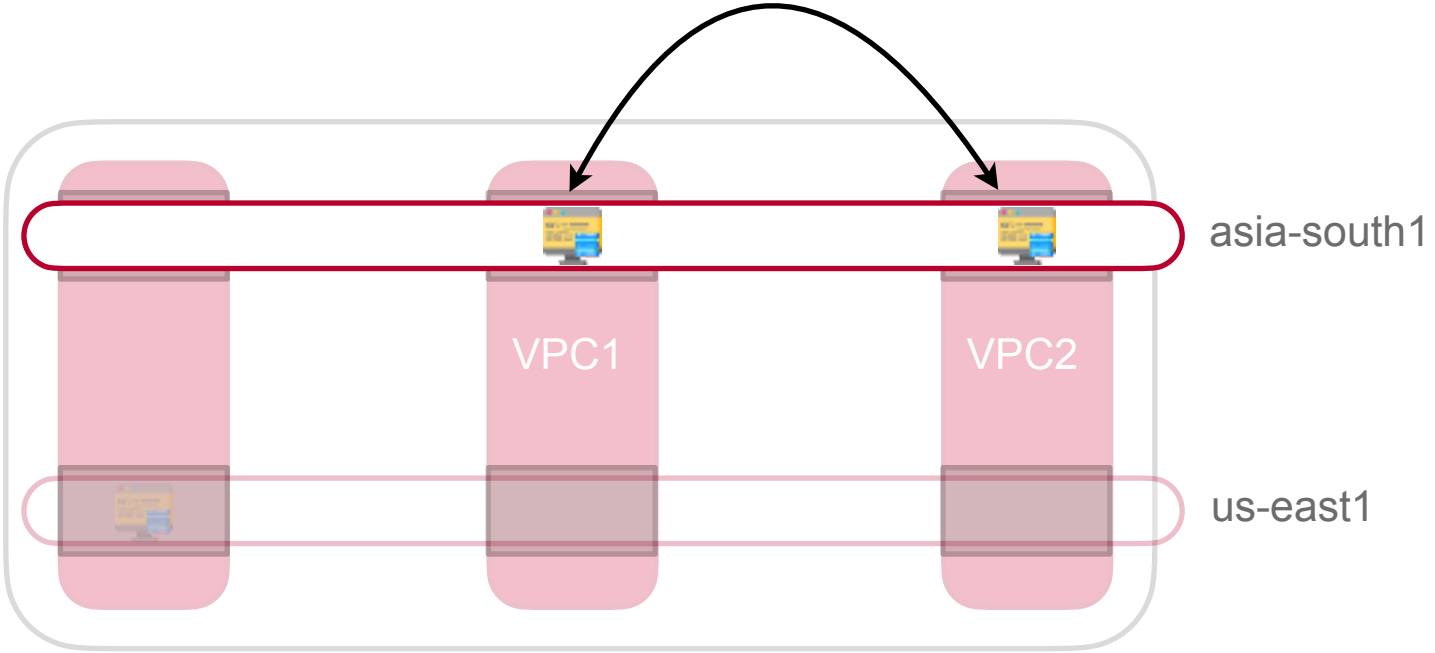
Wherever they are located in the world -
irrespective of physical location

Communication on VPCs



Resources on different VPCs communicate over the internet using external IPs

Communication on VPCs



Even though they are in the same region - they may even be in the same zone on the same physical hardware

Default VPC

- Pre-created on every project
- Includes subnet for each Google Cloud region
- New subnets added when new regions are created
- Resources created here by default



Default VPC

- Includes routes for all resources
- All VMs on the default VPC can talk to each other
- Default gateway to internet
- Includes several firewall rules



Firewall Rules

- Every VPC is a distributed firewall
- Firewall rules defined in VPC
- Are applied on per-instance basis
- Can also regulate internal traffic



Firewall Rules

- Every VPC has two permanent rules
  - Implied allow egress
  - Implied deny ingress
- Can be overridden by more specific rules
- In addition, default VPC has several rules



Additional Rules in Default VPC

- default-allow-internal
- default-allow-ssh
- default-allow-rdp
- Default-allow-icmp




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Hands on Demos - VPC Networks


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Connecting Networks

Shared VPC

- Share VPC across projects on GCP
- One VPC shared across projects
- Projects must be in the same organization
- Host project, guest resources
- Shared VPC admin to administer the shared VPC



VPC Peering

- Two or more VPCs shared across projects
- Projects need not be in the same organization
- Allows resources on different VPC networks to communicate using internal IP addresses
- Resources on the network use Google infrastructure to communicate
- Reduced latency, higher security and lower cost as compared with using external IPs



Shared VPCs vs. Network Peering

Shared VPCs

- Only within **same organization**
- One VPC used across projects
- Host and service projects not peers
- Only single level of sharing possible

Network Peering

- Across **organization boundaries**
- Multiple VPCs share resources
- Connected VPCs are peers
- Multiple levels of peering possible

Interconnecting Networks



GCP-to-GCP

VPC Network Peering

Enterprise connectivity

Peering and interconnect options

Interconnecting Networks

Enterprise connectivity

Peering and interconnect options

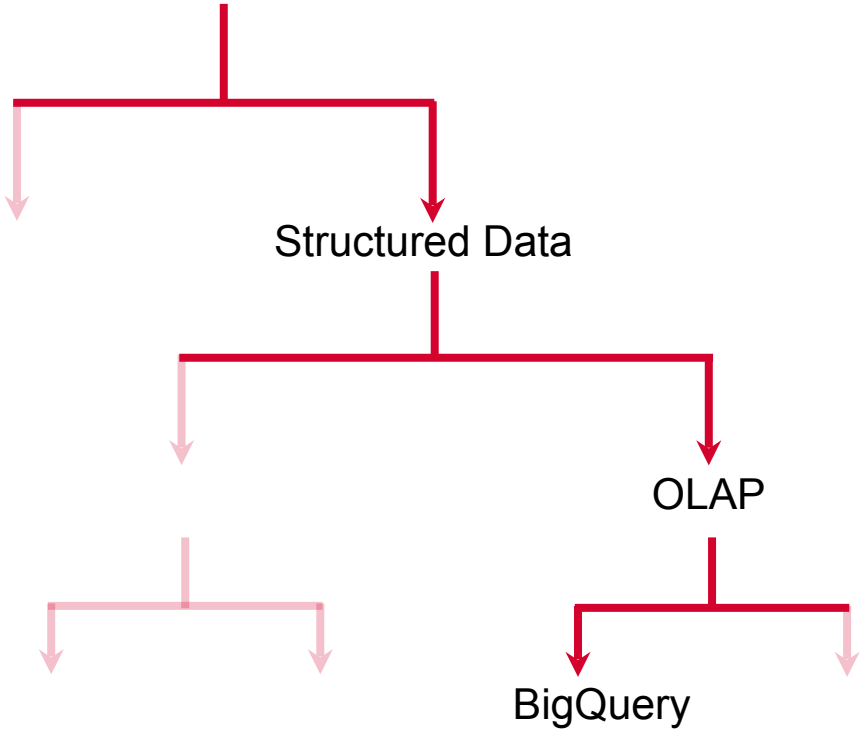
Connect a cloud network with an on-premise network using private or public IP addresses - VPNs, Cloud Interconnect

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BigQuery



Storage Technologies



BigQuery is a Data Warehouse that is hard to tell apart from an RDBMS

BigQuery vs. Traditional Data Warehouses



BigQuery

- Complex analytical queries
- Scales to Petabytes
- Both reads and updates
- Real-time or batch access
- Multiple data sources
- Streaming as well as batch

Traditional Data Warehouse

- Complex analytical queries
- Scales to Petabytes
- Mostly reads
- Long running jobs
- Multiple data sources
- Often more focus on batch

BigQuery Features

- Serverless: No cluster, no provisioning
- Autoscaling
- Automatic high availability



Support for the 4Vs

- Volume: Scales to Petabytes
- Variety: Federated data sources
  - Cloud storage
  - BigTable
  - Google Drive spreadsheets
- Velocity
  - Streaming ingestion
  - Real-time queries
- Variability
  - Schema auto-detection



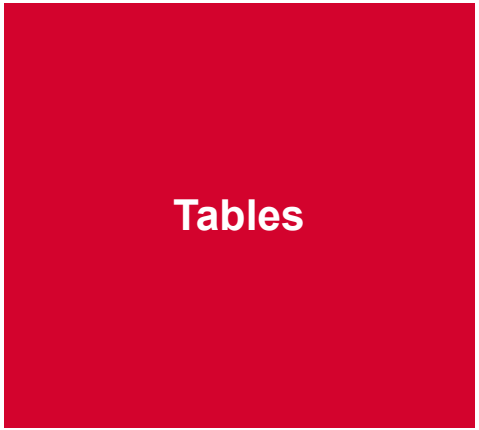
SQL Support

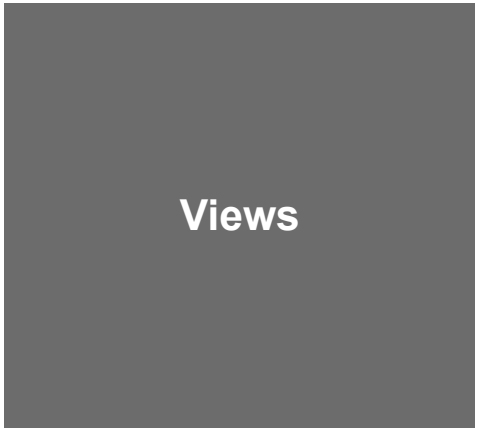
- Standard SQL
  - ANSI:2011 compliant
  - Extensions for nested/repeated fields



BigQuery







BigQuery Dataset

Top-level container used to organize and control access to tables and views. A table or view must belong to a dataset.

BigQuery Table



Contains individual records organized in rows. Each record is composed of columns (also called fields).

BigQuery View



Virtual table defined by a SQL query. Whenever a user queries the view, the underlying view-query is executed.

Advantages of Views

- Reduce query complexity
- Restrict access to data
- Construct different logical tables from the same physical table



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Hands on Demos - BigQuery



BigQuery

Which of the following statements about BigQuery is true?

1. BigQuery tops out after storing terabyte size data
2. BigQuery does not need cluster provisioning and set up
3. BigQuery offers transaction support at scale
4. BigQuery does not allow partitioning of data



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